

Adverse Selection Among Early Adopters and Unraveling Innovation

Rory M^cGee^{*}

September 17, 2025

Abstract

I analyse “selection markets”, where consumers differ in their willingness-to-pay and the costs they impose on sellers. Increased adverse selection among early adopters generates a dynamic free-rider problem—causing markets to unravel prematurely. With new products (e.g., reverse mortgages, insurance design) market power can mitigate inefficiencies stemming from adverse selection. Empirical evidence leverages differences in private information to show that early adopters in the market for deferred income annuities are more adversely selected. I then derive comparative statics ordering markets by their adverse selection, equilibrium prices, quantities, and welfare. These results explain three empirical phenomena: low uptake of existing products, slow demand for new products, and market inactivity despite unmet demand.

Keywords: Adverse selection, Insurance, Annuities; JEL Codes: D82, G22, D14

^{*}Department of Economics, University of Western Ontario and Institute for Fiscal Studies. Address: Department of Economics, Social Science Centre, University of Western Ontario, London, Ontario, N6A 5C2, Canada. Email: rmcgee4@uwo.ca. Website: <https://sites.google.com/view/rorymcgee/home>. I thank Roy Allen, Eric French, Nirav Mehta, Sergio Ocampo, Martin O’Connell, Cormac O’Dea, Jesse Perla, and Lisa Tarquinio for helpful comments and suggestions. Ian O’Donnell provided excellent research assistance. This research was supported by the University of Western Ontario Faculty Research Development Fund.

Selection markets—where consumers differ in the costs they impose on firms—are ubiquitous across annuities, insurance, credit, securities, used goods, and even labor markets.¹ These markets often exhibit low utilization, which empirical work typically attributes to low consumer demand.² However, this explanation overlooks a key puzzle: suppliers’ failure to offer more attractive products despite unmet demand. I show how adverse selection in immature markets exacerbates initial low demand and causes unraveling. Even markets that would be fully efficient at maturity fail to develop when early adopters are more adversely selected.

This paper’s central insight is that adverse selection in immature markets creates a novel dynamic free-rider problem. When early adopters are more adversely selected, markets may fail to mature because initial entrants bear the costs of adverse selection while subsequent entrants compete away long-run rents: preventing the cross-subsidization of initial losses. Consequently, initial adverse selection can cause permanent unraveling even when the mature market would be efficient.

This rests on whether early adopters are more adversely selected than consumers in mature markets—a theoretically ambiguous question. I formalize this argument with theory and evidence from the maturing deferred income annuity market. In immature markets, potential consumers have less confidence in products and derive lower benefits from adoption. As demand falls, selection patterns change: early adopters willing to buy despite lower initial valuations are precisely those imposing the highest costs on firms, making adverse selection generically higher in immature markets.

Using hand-collected annuity price data, I exploit age-based differences in available private information to separate adverse selection’s role from other strategic and price-relevant factors. I find economically and statistically significant decreases in adverse selection as the market matures. Groups with more private information experience larger equilibrium price reductions, consistent with theory. Taken together, this provides a *joint* explanation for three empirical regularities: low take-up of existing products, sluggish demand for new products, and market inactivity despite substantial unmet demand. For example, [Zinman \(2014\)](#) highlights missing rungs in consumer lending ladder, [Ameriks et al. \(2016\)](#) documents unmet demand for better long-term care insurance, and [Cocco and Lopes \(2015\)](#) establish demand for alternative reverse mortgage products. [Michaud and Amour \(2023\)](#) document potential demand for

¹[Einav et al. \(2021\)](#) provide an overview.

²See [Lockwood \(2012\)](#) and [Pashchenko \(2013\)](#) for annuities, [Ameriks et al. \(2016\)](#) and [Lockwood \(2018\)](#) in long-term care insurance, and [Nakajima and Telyukova \(2017\)](#) and [Cocco and Lopes \(2020\)](#) in reverse mortgage markets. Alternatively see [Einav et al. \(2010\)](#) for results emphasizing adverse selection.

bundled risk management products combining existing insurance. Yet take-up remains very low: Nakajima and Telyukova (2017) and Webb (2011) document poor adoption of novel reverse mortgage and annuity products, respectively. Similarly, despite being available since 1954, variable annuities (bundling tax-favorable saving, insurance, and indexed options) remained a small market until the early 1990s (Brown and Poterba 2006).³

The paper makes three important contributions. First, I outline this novel free-rider problem's implications in an intuitive framework. A key insight is that even would-be efficient markets can unravel during immaturity—a new, dynamic source of market failure building on Akerlof (1970). This free-rider problem challenges the common understanding of market power and adverse selection. Traditional static analyses, such as Starc (2014) and Mahoney and Weyl (2017), highlight that adverse selection can mitigate market power's negative impacts by constraining rent extraction, though market power exacerbates under-supply. However, dynamic analysis reveals an opposite mechanism: market power can alleviate dynamic inefficiencies by allowing monopolists to cross-subsidize early losses with longer-run gains, reversing policy implications.

Second, although testing hypothetical dynamics in nonexistent markets is impossible, I show the model predictions are empirically relevant using the market for deferred income annuities. I use a novel identification strategy to test for changes in adverse selection as the market matures using differences in the potential for private information. I find that adverse selection declines as the market matures and does so more strongly for consumers who purchase an annuity at older ages.

Third, I provide technical building blocks through general comparative statics results for selection markets under monotonicity assumptions, microfounding the payoffs that generate the dynamic free-rider problem. Since monotonicity is not guaranteed with multidimensional heterogeneity (Fang and Wu 2018), I characterize comparative statics using joint distributions of model primitives, developing an ordering over adversely selected markets by equilibrium quantity, prices, selection, and welfare. I provide explicit, economically interpretable sufficient conditions on joint distributions that characterize demand and cost curve shifts, accommodating various commonly used consumer payoff specifications. I show these same conditions

³Concern about missing insurance innovation dates back over half a century: Rudelius and Wood (1970) identify heterogeneous adoption of life insurance innovations, Dorfman (1971) provides evidence of research activity and barriers to innovation, and Murray (1976) surveys insurance companies to identify innovations.

are sufficient for graphical analysis of selection problems with multidimensional heterogeneity—a key contribution.

I establish these results in a standard insurance demand model with multidimensional heterogeneity. Consumers are heterogeneous across costs, risk premia (which determine insurance surplus), and other dimensions capturing frictions or departures from rationality. I consider adverse selection settings where consumers' willingness-to-pay correlates positively with their costs to insurers.⁴ To study unraveling in novel markets, I distinguish mature markets (well understood by consumers) from immature markets (new and unfamiliar), extending the framework by allowing consumers' willingness-to-pay to vary across market maturity.

These differences in components of willingness-to-pay capture objective factors, such as counterparty risk (Briggs et al. 2023) or self-insurance patterns consumers cannot quickly adjust, and subjective factors including distrust of new products, inertia (Handel 2013), or learning dynamics (Israel 2005). These differences lower demand and increase consumer sorting on costs, increasing adverse selection in immature markets and generating equilibrium under-insurance with lower adoption and higher costs.

Related Literature

I show that a novel free-rider problem manifests in the dynamics of adversely selected markets. Importantly, this reverses standard intuition for the interaction of market power and adverse selection. At any point in time, adverse selection impacts firm's ability to exploit market power as discussed by Starc (2014) and Mahoney and Weyl (2017). Moreover, market power exacerbates the under-supply driven by adverse selection. Here, however, the presence of market power early in a product's life cycle can incentivize entry when would-be efficient markets fail to mature.

This insight connects closely to Kong et al. (2024), who demonstrate that excessive price competition arising from “cream-skimming” incentives in adversely selected markets can undermine entry incentives and lead to market unraveling. This shows that market power can actually improve welfare in selection markets by preserving entry incentives. My analysis extends this fundamental insight to a dynamic setting, showing how the timing of competition matters crucially for market development. While Kong et al. focus on static equilibria where competition prevents profitable entry, I examine how the evolution of competition—with early entrants facing adverse selection but later entrants competing away rents—creates a dynamic free-rider problem that can prevent

⁴Analogous results can be obtained for advantageous selection.

markets from ever reaching maturity. The key distinction is temporal: early entrants must bear the costs of market development (including higher adverse selection) while later entrants can free-ride on this investment by entering only after the market matures. This temporal dimension generates distinct policy implications.

I examine the implications of changing adverse selection over time and study policies promoting initial market entry. These results emphasize that inter-temporal cross-subsidization can be optimal under missing markets. These policies are related to, but differ from, those designed to mitigate selection (Geruso and Layton 2017, review these policies in health insurance markets).

To microfound these dynamic incentives, I develop comparative static tools providing non-parametric conditions for ordering partial unraveling in Akerlof markets. This characterization allows theoretical insights to map onto commonly used empirical specifications, such as the constant absolute risk aversion utility in Cohen and Einav (2007). Building on Akerlof (1970)'s analysis of market unraveling, Hendren (2013) provides extended no-trade theorems for Rothschild and Stiglitz (1976) markets, while Attar et al. (2021) establish necessary and sufficient conditions for no-trade theorems in adverse selection markets.⁵ I extend non-entry results to cases where adverse selection declines over time, showing how these dynamics can lead to permanently missing markets even when static analysis would predict efficient outcomes

I study distributional changes in choice frictions similarly to Handel et al. (2019), providing comparative statics for non-local changes in both choice friction distributions and insurance value of contracts across markets. This builds a partial ordering of equilibrium prices, quantities, and welfare in terms of model primitives. Handel et al., however, study local reforms (including information interventions) that alter consumer's choice frictions and provide sufficient statistics formulas for their welfare impacts. As in Handel et al. and Spinnewijn (2017), I show how changes to model primitives lead to the rotations and shifts to cost and demand that are studied in other contexts (e.g. Starc 2014; Mahoney and Weyl 2017).

⁵A sizable literature studies endogenous contract terms under adverse selection (e.g. Rothschild and Stiglitz 1976; Handel et al. 2015; Veiga and Weyl 2016; Azevedo and Gottlieb 2017) or screening with intensive and extensive margins (Geruso et al. 2023), which is not this paper's focus.

1. Will Deep Pocketed Insurers Enter When Immature Markets Unravel?

I begin by studying how adverse selection among early adopters leads to would-be efficient markets failing to mature. To answer this, I illustrate the key economic forces determining insurer entry into new markets using a two firm, two period model. I show that adverse selection produces a novel form of dynamic free-rider problem which makes new markets difficult to grow.

I begin by studying how adverse selection among early adopters leads to would-be efficient markets failing to mature using a two-firm, two-period model that illustrates the key economic forces determining insurer entry into new markets. When adverse selection is severe enough, even the promise of market power cannot entice insurers to enter. Despite potentially efficient mature markets, *temporary and predictable* increases in adverse selection create missing markets, explaining the absence of better insurance products including redesigned long-term care insurance or reverse mortgage products.

These results reverse standard intuition about market power and adverse selection from static frameworks. In static analysis (e.g. [Starc 2014](#) or [Mahoney and Weyl 2017](#) when quantity is low), adverse selection ameliorates market power problems by limiting rent extraction, though market power exacerbates under-supply. This dynamic setting shows how market power can ameliorate dynamic inefficiencies by allowing monopolists to cross-subsidize early losses with longer-run gains—highlighting the complexity of policy intervention in selection markets.

In each period, firms choose whether to offer insurance taking contract terms as given. They observe entry decisions and post prices, competing à la Bertrand, so prices equal average costs if both firms enter. If only one firm enters, they exploit market power and equate marginal revenue with marginal cost. The market begins as immature and updates to mature in the second period if and only if at least one firm enters in the first period and sells positive quantity.⁶

The following result characterizes how market immaturity interacts with market forces and can endogenously create market incompleteness:

PROPOSITION 1. *If a monopolist cannot earn positive profits in immature markets, there exists a subgame perfect equilibrium where no firm enters. Moreover, this is the unique extensive-form trembling hand perfect equilibria.*

⁶Consumers learn about the generic "product" offered by firms instead of developing brand loyalty.

Importantly, this depends only on immature market primitives and holds even when the mature market is efficient. Static adverse selection in new markets can create missing mature markets regardless of mature market efficiency. The proof is in Appendix A.

The intuition reveals a new free-riding problem preventing market maturation. When early adopters are more adversely selected, a firm entering an immature market effectively invests (by earning negative profits) in market maturity.⁷ If they could guarantee monopolist profits in the mature market, absorbing temporary losses might be profitable. However, the entrant pays costs to reduce future adverse selection for all firms—by familiarizing consumers with new products—while competitors can wait and enter the mature market after this investment. Original entrants bear initial selection costs, but later entrants compete away potential long-run rents that could cross-subsidize initial losses. Consequently, no firm enters the immature market.⁸

This logic requires that no positive profits exist in the market’s infancy. Crucially, even monopolists have no guarantee of positive profits under adverse selection. Consider the standard profit margin expression (Starc 2014; Kong et al. 2024):

$$p - AC(p) = \frac{1}{\eta_{D,p}} - (AC(p) - MC(p)). \quad (1)$$

The first term is the usual Lerner markup and the second captures adverse selection’s influence through firms’ desire to “cream skim” low-cost consumers at the margin. This shows how adverse selection can discipline market power (Mahoney and Weyl 2017, note this effect reverses at high quantities). However, firms may earn negative monopolist profits absent fixed costs when either price elasticity of demand is large, $1/\eta_{D,p} \approx 0$, or the downward pressure of adverse selection is severe enough.⁹

Implications For Market Intervention: Patents or Burdensome Regulation as Time Varying Subsidies. If mature markets generate surplus, maturation may be socially optimal despite no-trade in infancy. However, permanent policy solutions, like a subsidy, may be inefficient because the primitives evolve over time. While missing mature markets create welfare costs, competitive allocations can be efficient once markets reach maturity. Phasing out subsidies when government intervention creates

⁷Indeed, at least one major Canadian auto-insurer offers individuals subsidies (above rating incentives) to enroll in new telematics contracts.

⁸This assumes aggregate state transitions only once contracts exceed a positive threshold, so firms cannot “enter” by charging prices guaranteeing zero profits and no sales.

⁹Kong et al., similarly discuss entry under imperfect competition and derive a threshold condition on the slope of the average cost curve to sustain a given market structure.

deadweight loss is optimal, but implementing optimal time-varying subsidies may be infeasible due to information requirements. Tautologically, there is no choice data if markets are missing.

A simple decentralized alternative is removing barriers to patenting or increasing patent protection length in insurance markets. This raises monopolist profits in immature markets and encourages entry by protecting against free-riders, offering an effective time-varying subsidy that depends only on patent length rather than complex subsidy schedules. However, information problems remain: planners need sufficient information to select optimal patent length.

Patents have advantages over subsidies because filing requires firms to disclose information, which can help determine optimal patent length.¹⁰ In contrast, planners relying on subsidies must will insurance products into existence, choosing from abstract menus of hypothetical ideas and hoping they are technologically feasible. Patent-based planners choose from observed innovations.

An alternative tool is increasing regulatory burden on insurers offering new products through extensive disclosure requirements or costly reporting mandates for new policies. This increases adoption costs and slows diffusion across firms (Benhabib et al. 2021), lengthening the period of market power.¹¹

2. A Model of Competitive Equilibrium in the Insurance Market

The previous section establishes that adverse selection creates a dynamic free-rider problem where market power can improve outcomes by allowing monopolists to cross-subsidize early losses with longer-run gains. I now show how evolving willingness-to-pay and consumer sorting generate the payoffs that create dynamic free-riding. I present a canonical model of insurance contracts. Subsequently, I discuss how differences between mature and immature markets lower demand and increase consumer sorting based on costs. I show this increases adverse selection in immature markets and generates under-insurance in equilibrium with lower adoption and higher costs, increasing adverse selection in immature markets and generating equilibrium under-insurance with lower adoption and higher costs.

¹⁰See Williams (2017) for a survey of other considerations in optimal patent design.

¹¹Empirical evidence highlights regulation's role in suppressing innovation (Griffith and Macartney 2014; Aghion et al. 2023).

2.1. Demand, Supply, and Adverse Selection

I consider an [Akerlof \(1970\)](#) model of insurance in the style of [Einav et al. \(2010\)](#) and [Einav and Finkelstein \(2011\)](#). A single perfectly competitive and risk-neutral insurer sells insurance to a unit continuum of heterogeneous individuals at a common price p . Contract terms are exogenous and common across all individuals.

Demand. Individuals are heterogeneous across risk type π_i which determines the expected cost to the insurer; their risk premium r_i ; and potential demand frictions ε_i . These are private information for individual i .¹² An individual buys insurance if and only if their *subjective* willingness-to-pay, defined as

$$\omega_i \equiv u(\pi_i, r_i, \varepsilon_i) = u(\underbrace{\pi_i + r_i + \varepsilon_i}_{\text{insurance value}}), \quad (2)$$

for any strictly increasing utility function $u(\cdot)$, exceeds the price of the insurance. In addition to their risk type, individual i 's risk premium, r_i , determines the surplus in the insurance market.

Following [Spinnewijn \(2017\)](#), individual choices depend on additively separable demand frictions, ε_i , which can distort insurance demand. This agnostic approach nests important features including borrowing constraints, inertia, subjective beliefs, and bounded rationality. These frictions are heterogeneous and can lead individuals to overvalue ($\varepsilon > 0$) or undervalue ($\varepsilon < 0$) the insurance contract. Since risk premia and frictions have identical effects on consumer choices, I focus on perceived risk premia net of behavioral influences $\tilde{r}_i \equiv r_i + \varepsilon_i$, for the majority of the analysis.

Demand is given by

$$D(p) = 1 - F_\omega(p) = \mathbf{1}[\omega > p], \quad (3)$$

where F_x denotes the CDF of variable x .¹³

Supply. The cost of providing insurance at price p is determined by the risk types of individuals purchasing insurance at that price.¹⁴ Consequently, *average* and *marginal*

¹²Conditioning on observable characteristics correlated with π_i may be illegal. Nevertheless, this can be interpreted as a model of insurance contracts in each partition of the observable characteristic space. This rules out screening contracts or changes to the screen – this may be particularly important in immature markets.

¹³Similarly, μ_x and σ_x^2 will denote the mean and variance of x 's distribution.

¹⁴I abstract from administrative loadings or fixed costs which don't affect the main result.

costs are given by:

$$AC(p) = E(\pi | \omega \geq p) \quad (4)$$

$$MC(p) = E(\pi | \omega = p) . \quad (5)$$

When willingness-to-pay, ω , increases in insurer cost, π , there is adverse selection; marginal costs increase in price and the MC curve is downward sloping.

Equilibrium. Assuming continuous, monotonic demand and cost curves ensures a unique equilibrium if it exists. The equilibrium price is given by the break even price that pools expected costs

$$p^* = p : D(p) = AC(p) . \quad (6)$$

Consequently, the equilibrium allocation may differ from an efficient allocation. An efficient allocation insures everyone whose willingness-to-pay exceeds their expected cost ($\omega_i > \pi_i$), those for whom the demand curve lies above the marginal cost curve (Panel A, Figure 1).

Furthermore, a positive equilibrium price does not necessarily exist. The insurance market unravels (Akerlof 1970) when individuals are not willing to purchase insurance at the pooled cost of higher-demand individuals. This occurs when the demand curve lies entirely below the average cost curve (Panel B, Figure 1 and Hendren 2013) .

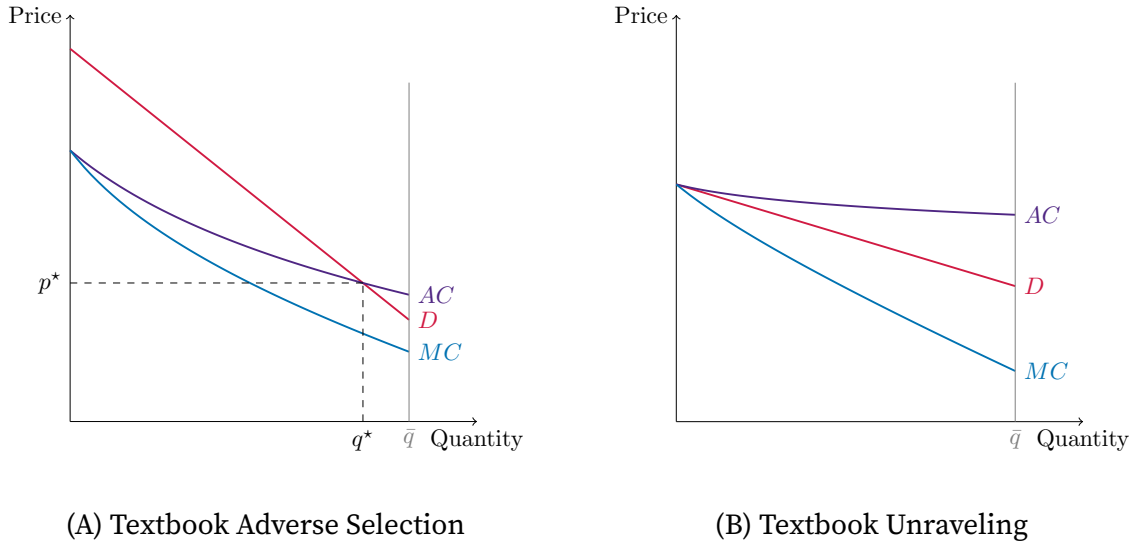


FIGURE 1. Equilibrium in the Insurance Market Without Demand Frictions

2.2. Comparative Statics Between Markets and Unraveling in Immature Markets

I now turn to the implications of market maturity for adverse selection and market unraveling. To do so graphically, I focus on cases where both demand and average and marginal cost curves are monotone. This standard restriction is widely used following [Einav et al. \(2010\)](#) and [Einav and Finkelstein \(2011\)](#) (see [Einav and Finkelstein 2023](#) for a recent review). Section 4 provides conditions on primitives that deliver monotonicity and states the assumptions on mature and immature markets in the general framework.¹⁵ Note that monotonicity does not require that choice frictions are independent of other characteristics ([Solomon \(2023\)](#) finds evidence in favour of dependence). Consequently, choice frictions are a reduced form compatible with a range of behavioral microfoundations including rational inattention models and correlated mistakes affecting both risk premia and insurance decisions ([Leive et al. 2022](#)).

To make markets comparable, I hold the distribution of potential and realized risks fixed across mature and immature markets: the product and risk pools are identical.¹⁶ I then assume willingness-to-pay is depressed in immature markets, i.e., demand contracts. This can be driven by objective risk premia, r , or from demand frictions, ε . I begin by motivating differences in the objective component of net premia, $\tilde{r}$.

Three forces generate objective differences between risk premia in immature versus mature markets. First, portfolios may not yet be optimized to take advantage of a new product (e.g., savings used for self-insurance crowding out formal insurance), so the benefits of insurance rise as other choices adjust. Second, there may be greater objective uncertainty about coverage, hassle costs, or reclassification risk; in health insurance, for example, coverage of new drugs/procedures is uncertain and heterogeneous, which depresses premia. This uncertainty is likely larger for new types of insurance and lowers the risk premia. Third, counterparty risk may be higher when insurers collect premia now but only begin paying out far in the future (e.g., life insurance, deferred annuities, long-term care), distorting insurers' portfolio choice ([Thompson 2010](#), models endogenous counterparty risk) and raising the likelihood of default or non-payment—again lowering risk premia.

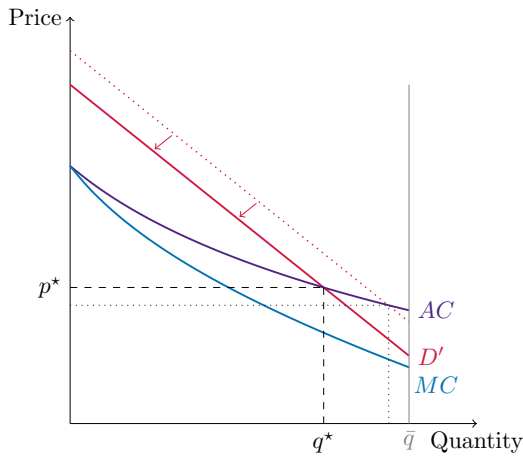
Alternatively, lower demand may reflect more negative demand frictions in

¹⁵These graphical results can be viewed as specific interpretation of the [Einav and Finkelstein](#) machinery to market maturity. However, the monotonicity assumption is non-trivial in multi-dimensions and to avoid repetition in Section 4 I present the graphical results as formal statements here.

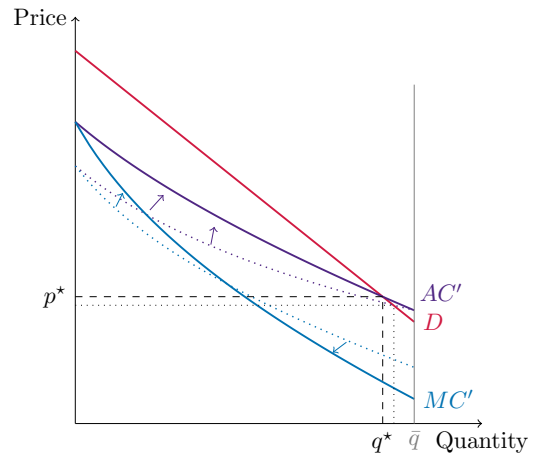
¹⁶Thus neither firms nor individuals can alter the risk distribution in mature markets.

immature markets: status quo bias, distrust of new products, inertia, lack of awareness (Boyer et al. 2020), or narrow consideration sets. Frictions that dissipate with learning or peer experience (Chetty et al. 2013; Duflo and Saez 2003) fit the same representation. In short, immature markets have *more negative* demand frictions than mature markets, where more consumers have prior exposure (peers, advertising, past purchase).

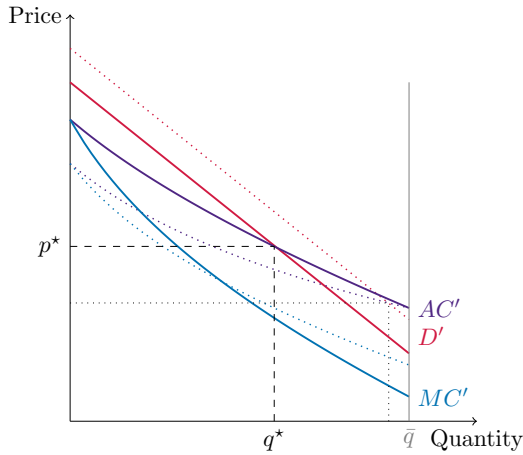
The implication is immediate: at any given price, average willingness-to-pay is higher in mature markets, so more individuals purchase insurance. Panel A of Figure 2 depicts this *level effect* on demand.



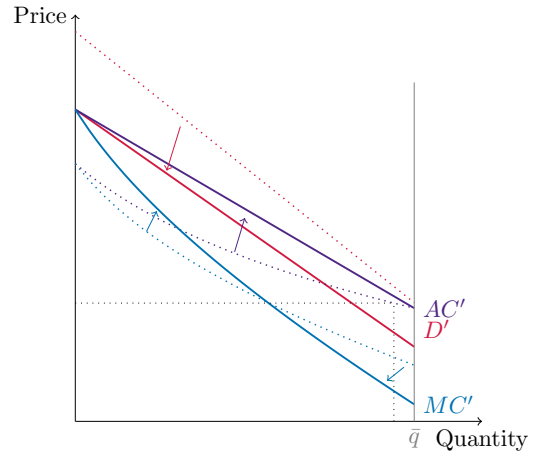
(A) Demand Contraction (Level Effect)



(B) Increased Adverse Selection (Sorting Effect)



(C) Under-insurance in the Short Run



(D) Unraveling in the Short Run

FIGURE 2. The Long and Short Run of Insurance Markets

This level effect holds average and marginal costs fixed at any given quantity. Although households require lower prices to induce purchase, the willingness-to-pay ordering is unchanged. Hence the identity of marginal (and inframarginal) buyers at a given quantity is the same, and equilibrium price rises while quantity falls (Panel A; Proposition 2).

Early adopters can also differ in sorting: beyond lower demand, they may sort more strongly on price,¹⁷ yielding a steeper marginal cost curve and more adverse selection.

Intuitively, the highest willingness-to-pay types may impose the highest coverage costs, making early adopters uniformly more costly on average. A pure level effect can raise average costs at a given price because it changes equilibrium quantity, but a change in sorting is required to shift cost curves *conditional on quantity*. When selection on cost increases, insurers face steeper marginal costs and higher average costs.

Order individuals by willingness-to-pay. Without re-sorting, this ordering is identical in mature and immature markets. Thus if q individuals buy insurance, the marginal consumer is the same and imposes identical expected costs on the insurer. With re-sorting, identities change: increased sorting on cost in immature markets implies that purchasing insurance signals higher average costs than in maturity.

Panel B of Figure 2 shows the outward expansion of the average cost curve in immature markets.¹⁸ Even holding the level of demand fixed, increased adverse selection among inframarginal consumers raises expected costs. Insurers set higher prices to cover these costs, despite some lower-cost buyers exiting. Equilibrium price rises and quantity falls: there is (additional) partial unraveling.¹⁹

The next result formalizes these implications of market maturity under monotonicity, demand contraction, and increased sorting.

PROPOSITION 2 (Partial Unraveling). *If an equilibrium exists, the equilibrium price is higher in immature markets and fewer consumers are insured, $q^{MM} \geq q^{IM}$ and $p^{MM} \leq p^{IM}$.*

Panel C of Figure 2 illustrates the new equilibrium. Marginal costs lie below demand in both mature and immature markets, so full insurance is efficient in both. Nevertheless,

¹⁷This is an empirical question. Section 3 provides an example where equilibrium supports increased sorting on price; Section 4 formalizes the assumption.

¹⁸This is a rotation around full insurance as in Mahoney and Weyl (2017). The difference in average costs in (21) is decreasing in quantity, reaching zero at full coverage: $AC^{IM}(1) = AC^{MM}(1) = \mu_\pi$. In contrast, Starc (2014) considers rotation around equilibrium demand.

¹⁹This can occur even if marginal costs in immature markets lie below demand which, absent choice frictions, would make full coverage efficient.

relative to the mature market there is additional under-insurance due to both the level effect and increased adverse selection in the immature market. The equilibrium price increase and quantity reduction exceed either effect in isolation (Panels A and B).

Immaturity can also cause complete unraveling.

COROLLARY 1 (Complete Unraveling). *Would-be efficient mature markets can unravel before they reach maturity.*

This happens when the average cost curve lies above demand for all prices (Panel D of Figure 2). Thus, even a market that would be efficient if it could mature can unravel in its early stages. As Section 1 shows, this can lead to permanently missing markets.

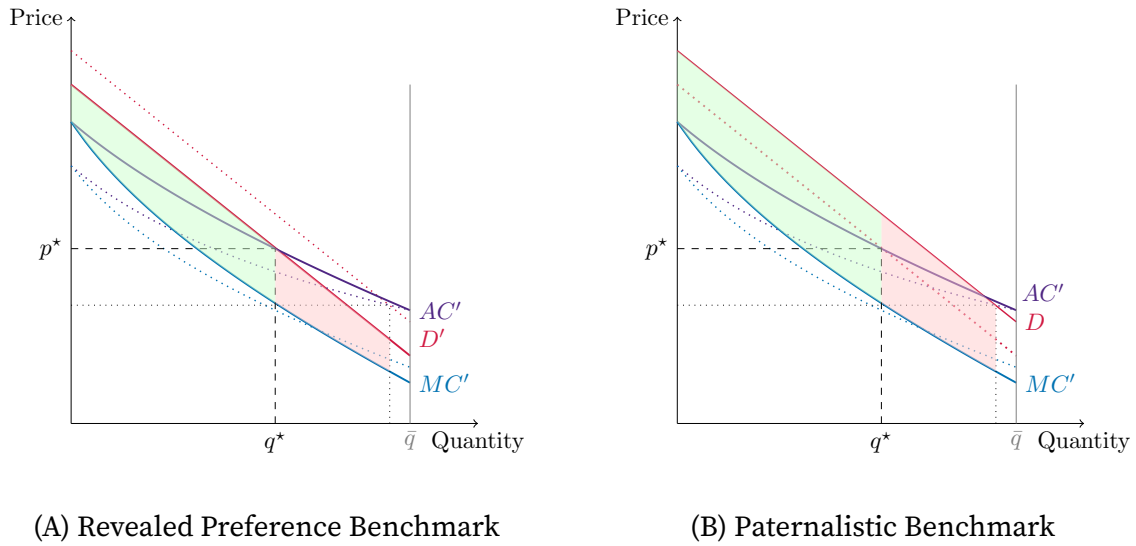


FIGURE 3. Welfare Losses in Immature Insurance Markets

Under-insurance ($q^{IM} < q^{MM}$) does not depend on whether lower willingness-to-pay is driven by risk premia or choice frictions. Computing welfare costs, however, requires a view on their relative importance. The next proposition evaluates welfare under two policy-relevant benchmarks, illustrated in Figure 3.

PROPOSITION 3 (Efficiency Costs). *Consumer surplus in immature markets is lower at the immature market equilibrium than the mature market equilibrium ($CS^{IM}(q^{IM}) \leq CS^{IM}(q^{MM})$). This is true irrespective of whether demand for insurance is evaluated at:*

- i. *Willingness-to-pay in immature markets (revealed preference benchmark)*

ii. *Willingness-to-pay in mature markets (paternalistic benchmark)*

Thus, the equilibrium in immature markets is pareto dominated by the mature market.

3. Identifying Adverse Selection Among Early Adopters: Evidence from the Market for Deferred Income Annuities

I now provide evidence that early adopters are more adversely selected. As there is no data on missing markets, I instead study the comparison between different stages of market maturity and show how they align with testable implications outlined above. Specifically, I focus on deferred income annuities – a product similar to a traditional annuity, but where payouts do not begin until a date (or age) agreed in the future. Hence, they provide deferred income.

Data on the aggregate size of the market, as measured by the dollar value of new annuity considerations, is collected from the quarterly reports of the Life Insurance Marketing and Research Association.²⁰ Figure 4A shows how cumulative sales evolved over time. At the start of this period the market for deferred income annuities was small, totaling around \$200 million and accounting for less than 1% of the overall fixed income annuity market (Panel B). The market expanded rapidly between 2012 and 2014, with the real value of new considerations more than quadrupling in size.

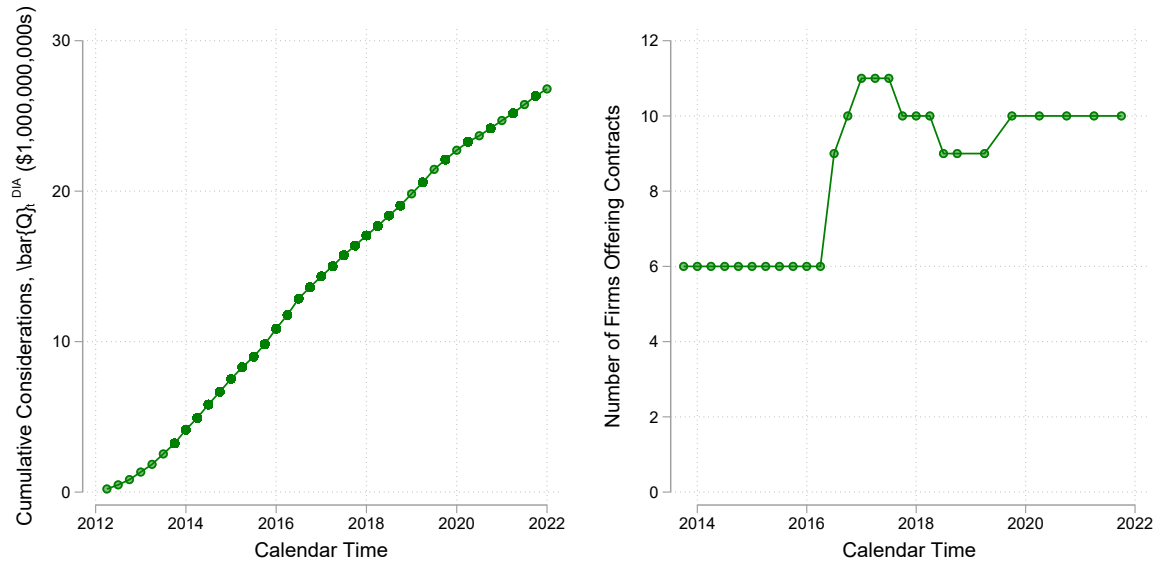
To study the equilibrium relation between price, quantity, and selection I complement this data on aggregate quantities with hand-collected quotes for individual annuity policies sourced from *Annuity Shopper* magazine between 2013 and 2021.²¹ The *Annuity Shopper* magazine was a longstanding product in the annuity and life insurance industry, targeting potential annuitants as well as financial advisors and Human Resources professionals. Consequently, it is in the interests of life insurers to provide quotes on their products.²²

In each period, national insurance companies provide quotes for a slate of policies. I collect a detailed quarterly panel dataset of the annuity payouts offered by insurers for homogeneous contract terms with 5,882 observations. These policies vary in the payouts they offer, the life insurer who provides the annuity, the age at which the annuity begins

²⁰Specifically, the U.S. Individual Annuity Sales Survey published by the LIMRA Secure Retirement Institute. Sales are not reported pre-2012 as the size of the deferred income annuities market is minimal.

²¹Previously The Annuity Shopper's Buyers Guide. Before the third quarter of 2013 quotes for deferred income annuities are not available and, sadly for retirement researchers, it ceased to operate in 2021.

²²Indeed, over 40 large US insurers provided quotes for a variety of products in the July 2015 issue. Appendix C provides an example testimonial.



(A) Cumulative Deferred Income Annuities Considerations (B) Distinct Firms Operating in the Market for Deferred Income Annuities

FIGURE 4. Growth In The Deferred Income Annuities Market

to provide an income stream, as well as the current age of the purchaser and their gender. In each period there are up to 28 different quotes available for each insurer varying in their contract terms and I focus on single life non-refundable policies without a guarantee period (Einav et al. 2010, study adverse selection in the guarantee decision). An important advantage of this empirical setting is that the product is homogeneous. This differs from health insurance, a common empirical setting, where differences in in- and out-of-network providers lead to differentiated products.

Insurers price annuities for men and women separately, reflecting differences in their survival risk. In addition, annuities are priced separately by the age at which the consumer is purchasing the annuity contract and the the deferral age at which the contract begins paying out. Quotes are available for three purchase ages (45, 55 and 60) as well as 5 different deferral ages (65, 70, 75, 80, and 85). Given policy characteristic at a point in time, quotes provide the annual payout offered per \$100,000 paid for the deferred income annuity.

Figure 4B shows how the number of firms offering deferred income annuities changes over time. Broadly, there are two regimes: first, a period where the number of entrants is stable at six distinct insurance firms; and second, a period averaging 10 firms. This expansion in the number of operating firms coincides with reforms to the fiduciary duty of brokers operating in brokerage markets (Egan et al. 2022).

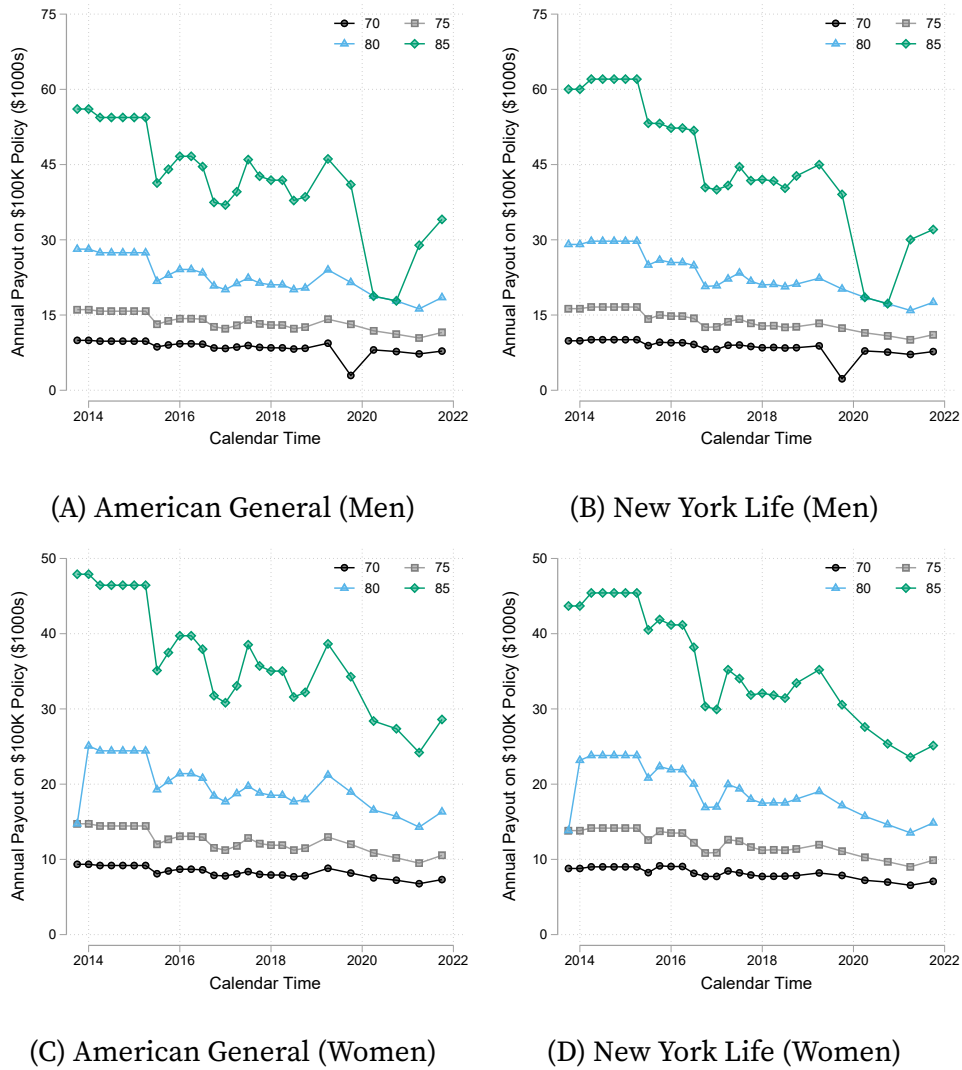


FIGURE 5. Average Payouts from Deferred Income Annuity Policies Purchased at Age 65 By Insurer, Gender and Deferral Age

Figure 5 shows payouts from annuities purchased at the age of 65 with policies offered by American General shown in the left column and policies offered by New York Life shown in the right column, two of the largest national insurers. The first row shows policies for men and the second row shows policies for women with each line corresponding to a deferral age from age 70 to 85. Three important facts emerge from this comparison. First, there are clear differences in policy generosity by gender. Second, there are systematic differences by the age at which the contract begins paying out. Policies with later deferral dates, which expose consumers to greater mortality risk, offer larger payouts on average and systematic differences across men and women are

consistent with differences in their expected longevity. This highlights the potential for selection to drive pricing decisions by firms in this market. Finally, comparing across the two columns, there are small deviations in the payout by insurer.

Figure A2 in Appendix C repeats this exercise for contracts offered by New York Life, but conditioning on the deferral age and varying the purchase age. Again, contracts that offer payouts further into the future have higher payout amounts. Despite systematic differences by product characteristics, a notable feature of both Figures 5 and A2 is evidence of a market level trend in payouts over time for all contract terms and for both insurers. This reflects that the largest cost to the insurer is the opportunity cost of foregone investment (see Mitchell et al. 1999) as well as a changing pool of annuitants.

I now turn to measuring selection and maturity, the key quantities of interest, before describing the empirical strategy used to test the implications of the model in Section 2.

3.1. Measuring Selection and Market Maturity

Each policy offered to annuitants at a purchase age, a , of gender, g , is indexed by the deferral age, d , at which payouts begin, the identity of the insurance firm, i , as well as the date, t , at which it is purchased. Each annuity, $\{a, d, g, i, t\}$, offers a fixed income A after the deferral age. I follow Finkelstein and Poterba (2002) and adopt a money's worth calculation to infer selection from annuity payouts. To do this, I calculate the expected present discounted value of the associated annuity income stream

$$EPDV_{a,d,g,i,t} = \frac{1}{1 - \chi_a(g, t - a)} \sum_{h=a}^{110} \frac{A \times \mathbb{1}[h \geq d] \times \chi_h(g, t - a)}{\prod_{j=h-a}^{110-a} (1 + i_j(t))}, \quad (7)$$

where $\chi_h(g, t - a)$ is probability of surviving until age h as a function of annuitant gender and birth cohort $t - a$. The insurer's interest rate over a horizon j from date t is $i_j(t)$. From this I compute the money's worth value of the annuity (its implicit price):

$$MW_{a,d,g,i,t} = \frac{1}{100,000} EPDV_{a,d,g,i,t}. \quad (8)$$

A money's worth value less than one indicates that the annuity is expected to pay out less than the upfront cost and vice-versa. In the absence of annuity product specific mortality tables (see Finkelstein and Poterba 2004 for evidence of differential mortality across products) the money's worth value can be used to infer deviations from the mortality table used to calculate it. In this sense, it allows researchers to infer differential

mortality and the presence of adverse selection (see also [Warshawsky 1988](#); [Mitchell et al. 1999](#); [Poterba and Solomon 2021](#)).²³ I build a measure of selection using the per dollar deviation from the expected present discounted value under zero adverse selection using population mortality tables.

I use the BBB corporate bond yield at 10-year maturity to determine the insurer's interest rate, i_j , and cohort life tables for the age a population to determine mortality risk $S_h(g)$ at a given point in time (linearly interpolating between successive cohort life tables). The deviation from actuarially fair pricing provides a measure of implicit price and implied adverse selection for each contract. Using gender-specific mortality to calculate money's worth, the implied adverse selection is given by

$$S_{a,d,g,i,t} = 1 - MW_{a,d,g,i,t}, \quad (9)$$

where a larger positive value indicates greater adverse selection.

The testable implications of Section 2 relate adverse selection and market maturity. I proxy unobserved market maturity using the natural logarithm of cumulative sales in the market for deferred income annuities up to that point in time (Figure 4A), $\ln Q_t^{DIA}$. As the measure of new considerations used to construct cumulative sales is not available at a finer disaggregation I use total sales for all deferred income annuities, rather than sales at the characteristic level. This has an intuitive interpretation as a measure of broad market popularity that corresponds to maturity discussed in Section 2. Using a log transformation implies that there is a concave-increasing relationship between maturity and cumulative sales of deferred income annuities and allows for an elasticity interpretation of the relationship between adverse selection and market maturity.

3.2. Empirical Strategy

Given this measure of adverse selection for each contract and a suitable proxy for market maturity, I now describe the empirical strategy used to test the model predictions. A key threat to identifying the effect on adverse selection is that alternative strategic considerations (e.g., dynamic price discrimination) in maturing markets may generate observationally equivalent pricing patterns even absent adverse selection. Thus the correlation between maturity on pricing may not capture the effect of adverse selection.

Absent private information, it would not be possible for adverse selection to change

²³[Cannon and Tonks \(2016\)](#) argue that prudent insurers may deviate from actuarially fair pricing when future mortality risk is uncertain. The empirical strategy below allows for this.

as the market evolves. Therefore, to address these identification concerns and directly test the model's implications, I use differences in available private information by age to exploit variation in the potential for adverse selection. Specifically, as health inequality increases by purchase age (see, e.g., [Hosseini et al. 2022](#); [Russo et al. 2024](#)), private information about the future mortality of potential annuitants is also increasing. Proposition 6 in the general model below formalizes this simple intuition.

This motivates an empirical strategy that exploits differences in the effects of market maturity by consumer's purchase age. I leverage the differential effect of market maturity between those age 45 and those aged 55 and 65. The key identifying assumption is a form of parallel trends assumption: (conditional on other controls) strategic considerations, other pricing decisions, administrative costs, profit rates, and other price-relevant factors would evolve similarly with market maturity if 55 and 65 year olds had the same level of mortality relevant private information as 45 year old consumers. This strategy is implemented by estimating the following specification

$$\ln S_{a,d,g,i,t} = \gamma_a + \gamma_d + \gamma_i + \gamma_g + \lambda \ln q_t^{FIA} + \beta \mathbb{1}[a = 45] \times \ln Q_t^{DIA} + \delta \mathbb{1}[a > 45] \times \ln Q_t^{DIA} + u_{a,d,g,i,t}, \quad (10)$$

where γ_a denotes purchase age specific fixed effects, γ_d denotes deferral age specific fixed effects and, γ_g denotes gender specific fixed effects. Collectively, these fixed effects capture systematic differences in selection or administrative costs by contractible plan characteristics that do not vary over time. β captures how pricing evolves with market maturity for age 45 purchasers and δ captures the differential effect for groups with more private information. This is akin to a parametric difference-in-differences estimator as it exploits differences in the within purchase age selection as the market matures. In this analogy, 'treatment' is determined by ex-ante difference in exposure (private information) and estimation uses continuous variation in market maturity.

The parameter of interest is δ which measures the differential effect of market maturity on pricing for groups with more private information (and, hence, a greater propensity for adverse selection). A key testable implication of the model outlined above is that adverse selection declines as markets mature. This corresponds to $\delta < 0$.

As the market matures, other aggregate factors may simultaneously drive differences in the choices of annuitants or the composition of the pool of potential annuitants including population level mortality improvements, differences in the financial resources of retirees over time, or the effect of legislative reform (such as the

change to broker fiduciary duty studied by Egan et al. 2022). These factors may be correlated with a positive time trend in market maturity and bias the estimate of maturity on adverse selection. To minimise this threat to identification, I control for contemporaneous demand in other fixed income annuity products excluding deferred income annuity considerations (Figure A1, Appendix C), $\ln q_t^{FIA}$, which absorbs time varying shocks affecting all fixed income annuity products.

Similarly, changes in average prices over time may reflect changes in the composition of insurers due to entry and exit. If insurers serve different populations due to heterogeneous geographic penetration or face different administrative costs, this would confound the effect of market maturity. I control for insurer-specific fixed effects which accounts for compositional changes in firms.²⁴

In addition to the specification in equation (10), I also estimate the following:

$$\ln S_{a,d,g,i,t} = \gamma_a + \gamma_d + \gamma_i + \gamma_g + \lambda \ln q_t^{FIA} + \beta \mathbb{1}[a = 45] \times \ln Q_t^{DIA} + \delta_{55} \mathbb{1}[a = 55] \times \ln Q_t^{DIA} + \delta_{65} \mathbb{1}[a = 65] \times \ln Q_t^{DIA} + u_{a,d,g,i,t}. \quad (11)$$

This differs in explicitly examining the heterogeneity between consumers who purchase annuities at age 55 and those who purchase at age 65. Examining this heterogeneity allows for a direct empirical test of a second testable implication, that markets with greater (cost-relevant) private information experience larger reductions in adverse selection as markets mature ($\delta_{65} > \delta_{55} > 0$).

3.3. Adverse Selection Among Early Adopters

Table 1 reports the results from estimating equation (10). The first column only controls for fixed differences across purchase ages. The estimated effect of additional private information on the elasticity of selection (as a deviation from actuarially fair pricing) with respect to market maturity is negative. A 1% increase in past cumulative sales of deferred income annuity products is associated with a -0.191 percentage reduction in selection over and above strategic price dynamics for purchase ages with less private information. Thus, as markets mature adverse selection decreases. This is direct evidence of larger adverse selection among early adopters.

The estimated effect is economically significant and statistically significant at the

²⁴Although annuity are fairly standardized. Insurers may differ in other ways such as customer service or integration with other financial services. The insurer fixed effect also controls for time invariant product differentiation. Section 4.4 extends the theoretical predictions to this setting.

	(1)	(2)	(3)	(4)	(5)
$\mathbb{1}[a > 45] \ln Q_t$	-0.191*** (0.034)	-0.224*** (0.030)	-0.227*** (0.028)	-0.235*** (0.029)	-0.238*** (0.027)
<i>FEs</i>					
Gender	No	Yes	Yes	Yes	Yes
Deferral Age	No	Yes	Yes	Yes	Yes
Insurer	No	No	No	Yes	Yes
FIA Demand	No	No	Yes	No	Yes
N	5, 857	5, 857	5, 857	5, 857	5, 857
R ²	0.110	0.409	0.486	0.440	0.514

Notes: * $p < 0.10$, ** $p < 0.5$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient δ in equation (10). Implied selection is recovered using equation (9) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators.

TABLE 1. Adverse Selection and Market Maturity

1% level; rejecting the null hypothesis of no or advantageous selection. Consistent with the identification argument, I also find that measured selection is on average larger for older purchase ages. Nevertheless, the way in which it *evolves* with maturity drives the estimated effects reported here.

The second column adds additional controls for important contract characteristics: gender and deferral age. As the pricing of annuities varies across these characteristics there is a large increase in the overall explanatory power as measured by the R-squared. However, there is only a small impact on the estimated decline in adverse selection. The elasticity increases, but is overall similar to the previous point estimate. This is because these plan characteristics deliver systematically different pools of annuitants, but evolve similarly over time.²⁵ The third column adds observed demand for all other fixed income annuities, $\ln q_t^{FIA}$, a proxy for aggregate factors that drive annuity demand and pricing in general. Although this absorbs some of the common effect of market

²⁵Estimated effects are qualitatively similar when estimated separately for each gender and deferral age grouping exploiting the same variation in private information across purchase ages. In some cases, these results have less precision (as each subsample is approximately one-tenth of the pooled sample).

maturity on pricing, it does not affect the point estimate because the empirical strategy exploits differences across purchase ages to identify the component driven by selection.

The fourth column instead includes insurer specific fixed effects to control for changes in the composition of insurers who may face different pools of annuitants or administrative costs. Overall, the point estimate increases slightly although the differences across columns three and four are statistically insignificant. This reflects the fact that these are national insurers who cover the whole US population and, thus, face similar pools of potential annuitants. Although there are differences in administrative costs across insurers (Mitchell et al. 1999; Kojen and Yogo 2015), these are, on average, only weakly correlated with entry into this market. The final column combines the controls in columns four and five. Across all sets of controls, the estimated elasticity are stable in magnitude and statistically significant at the 1% level. The empirical evidence in this market highlights that adverse selection can be higher among early adopters.

	(1)	(2)	(3)	(4)	(5)
$\mathbb{1}[a = 55] \ln Q_t$	-0.126*** (0.037)	-0.158*** (0.031)	-0.161*** (0.029)	-0.168*** (0.031)	-0.171*** (0.029)
$\mathbb{1}[a = 65] \ln Q_t$	-0.273*** (0.035)	-0.307*** (0.030)	-0.311*** (0.028)	-0.320*** (0.029)	-0.323*** (0.027)
<i>FEs</i>					
Gender	No	Yes	Yes	Yes	Yes
Deferral Age	No	Yes	Yes	Yes	Yes
Insurer	No	No	No	Yes	Yes
FIA Demand	No	No	Yes	No	Yes
N	5,857	5,857	5,857	5,857	5,857
R ²	0.115	0.414	0.490	0.445	0.519

Notes: * $p < 0.10$, ** $p < 0.5$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient δ_{55} and δ_{65} in equation (11). Implied selection is recovered using equation (9) and panel data collected from policy quotes printed in *Annuity Shopper* magazine.

Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators.

TABLE 2. Adverse Selection and Market Maturity by private information

Table 2 reports the results of estimating the specification in equation (11). The key difference is that this disaggregates the elasticity into separate estimates for age 55 and 65 purchasers. As with the results in Table 1, the parameter estimates differ slightly when controls for contract characteristics are added (columns one compared with two). Again, the point estimates are largely stable across alternative controls.

The first row reports the elasticity of selection for age 55 purchasers. These estimates vary between -0.126 and -0.171 and are consistently negative. The second row reports the elasticity of selection for age 65 purchasers. These estimates are larger in magnitude as the pooled effect is the weighted average, varying between -0.273 and -0.323 , and are also consistently negative. Like the pooled elasticity they are always statistically significant at the 1% level and reject the null hypothesis of no or advantageous selection in each age group.

Moreover, the disaggregated elasticities allow for a test of the second prediction: all else equal, early adopters are more adversely selected when there is more private information. Estimated elasticities increase with purchase age (and therefore private information) and their difference is negative and statistically significant. This directly tests the second testable implication and also lends support to the identifying assumption of less private information for younger age groups.

Robustness and Alternative Mechanisms. The empirical evidence presented here uses differences in the availability of private information by purchase age to identify the dynamics of adverse selection in a maturing market and is consistent with testable implications from theoretical analysis above. In Appendix C I provide further results and probe into the robustness of these results. In particular, I show that these results are robust to alternative controls for common aggregate factors as well as a more granular set of controls for policy characteristics. Moreover, the qualitative insights are robust to alternative specifications of the dependent variable including selection in levels and calculating implied selection using alternative mortality tables.

Nevertheless, it may be the case that alternative mechanisms produce these heterogeneous dynamics by purchase age. I briefly discuss two potential mechanisms and the reasons that they are unlikely to be responsible for the dynamics in this market.

First, in a new market firms may be learning about the distribution of risks (cost types) and may do so differentially among the pools of potential annuitants at different purchase ages. This could lead to initial miss-pricing and induce a negative correlation. However, although deferred income annuities are a new product, insurers are

experienced in pricing annuity and life-insurance contracts based on the distribution of underlying mortality risk. As equation (9) shows, the actuarial formula for a deferred income annuity payout is almost identical to the formulas for these existing products. They differ only in the timing at which payout begins. Therefore, it would have to be the case that insurers were learning about the declining adverse selection by age as the market matures – precisely the model’s prediction.

Second, firms may initially ‘invest’ in market power, e.g. by undercutting or offering ‘sweetheart’ deals, and subsequently raise markups to ‘harvest’ downward sloping residual demand.²⁶ This would produce a negative contemporaneous correlation between equilibrium prices and demand, a distinct implication to the time path considered here. Additionally, this mechanism has sharp predictions about the declining number of insurers in the market which is at odds with Figure 4. The number of insurers offering these contracts grows over time – inconsistent with price and quantity dynamics driven by this form of strategic behaviour.

4. Generalized Problem

The implications derived in Section 2 rely on monotonicity. As Fang and Wu (2018) note, selection problems with multidimensional types do not generically satisfy monotonicity.²⁷ I now provide sufficient conditions for monotonic cost curves and adverse selection under arbitrary marginals – a prerequisite for unraveling. Subsequently, I formalize how characteristics of immature and mature markets affect this unraveling.

4.1. Sufficient Conditions for Monotonicity and Adverse Selection

With multidimensional heterogeneity, the critical assumption for adverse selection is this dependence between individual willingness-to-pay and costs (Fang and Wu 2018). This depends on the copula representing their joint distribution.²⁸ Copulas encode the correlational structure between π_i , r_i , and ε_i without imposing parametric families,

²⁶This may arise if consumers do not switch products due to search costs. This is unlikely to be of relevance here where the annuitization decision is typically once and for all.

²⁷This motivates Fang and Wu to consider local adverse (and advantageous) selection in a multidimensional context. Their Definition 2 corresponds to the implication of component-wise concavity in Proposition 4 below and the global conditions have a natural analogue for monotone regions and local adverse selection.

²⁸Copulas have many uses in economics and finance – particularly in the context of selection and dependent risks (Kim 2021; Gola 2021).

decoupling dependence assumptions from marginal distributions.²⁹ They encode this structure as as rank dependence restrictions in quantile space (Anderson and Smith 2024), allowing *sorting* to be disentangled from cost and risk premia distributions.

Although results can be characterized for the special case of joint-normality, this restricts the risks consumers experience and insurers insure, ruling out heavy-tails and skew important in actuarial modeling where tail events significantly impact balance sheets despite infrequency. Consider the following example motivated by the empirical specification in Cohen and Einav (2007) for an illustration of the limits of normality: consumers have constant absolute risk aversion θ , and face a financial loss m which arrives with probability λ . Then

$$\begin{aligned} \text{Certainty Equivalent:} \quad CE_i &= \frac{\ln(1 - \lambda_i + \lambda_i e^{-\theta_i m})}{\theta_i} \\ \text{Expected Cost:} \quad \pi_i &= \lambda_i m \\ \text{Risk Premia:} \quad r_i &= \frac{\ln(1 - \lambda_i + \lambda_i e^{-\theta_i m})}{\theta_i} - \lambda_i m . \end{aligned}$$

When risk aversion and the the arrival rate vary across individuals the distribution of risk premia is non-normal even when λ and θ are (log-)normally distributed. Although copulas increase complexity, they allow the framework to describe economically and empirically relevant settings.

The following proposition provides sufficient conditions on joint distributions for monotonicity without marginal distribution restrictions using the familiar monotone likelihood ratio property (Milgrom 1981):

PROPOSITION 4 (Adversely Selected Markets). *Let the density $f(\cdot)$ satisfy the Monotone Likelihood Ratio Property (MLRP), i.e. $f(\omega_1|\pi_1)/f(\omega_1|\pi_2) \geq f(\omega_2|\pi_1)/f(\omega_2|\pi_2)$ for any $\omega_1 \geq \omega_2$ and $\pi_1 \geq \pi_2$, then:*

- i. The market is adversely selected and marginal costs, $MC(p)$, is increasing in price.*

Sklar's Theorem: Let F be a joint distribution function for a d -dimensional random vector $(X_1, X_2, \dots, X_d)$, and let F_i be the marginal distribution function for the i -th component X_i . Then, there exists a copula C such that for any $x_1, x_2, \dots, x_d$ in the real line:

$$F(x_1, x_2, \dots, x_d) = C(F_1(x_1), F_2(x_2), \dots, F_d(x_d))$$

Moreover, if X is continuous, then the Copula C is unique.

²⁹Copulas are invariant to rank-preserving transformations, so results immediately apply to multiplicatively separable utility, $u = \exp(\cdot)$, and when insurer costs are transformed cost types.

- ii. Average costs, $AC(p)$, are monotonically increasing in price.
- iii. The adverse selection wedge, $AC(p) - MC(p)$, is positive.

The formal proof is given in Appendix D. Intuitively, the MLRP property allows the insurer to infer higher costs from the signal of higher willingness-to-pay. Crucially, this does not depend on marginal distributions. For a bivariate distribution with copula $C(u, v)$ and copula density $c(u, v)$, MLRP holds if and only if:

$$\frac{\partial^2}{\partial u \partial v} \log c(u, v) \geq 0. \quad (12)$$

This restricts the shape of the scale-invariant rank dependence *within* market, formalizing that larger X values tend to be realised with larger Y values. It implies positive quadrant dependence and positive correlations and regression coefficients.³⁰

For joint-normality, π and $\tilde{r}$ are multivariate normal with correlation $\rho_{\pi, \tilde{r}}$, this is a restriction on the correlation and the joint distribution of cost type and willingness-to-pay is given by,

$$\begin{pmatrix} \pi_i \\ \omega_i \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_\pi \\ \mu_\omega \end{pmatrix}, \begin{pmatrix} \sigma_\pi^2 & \rho \sigma_\pi \sigma_\omega \\ \rho \sigma_\pi \sigma_\omega & \sigma_\omega^2 \end{pmatrix} \right],$$

where $\rho = \frac{\sigma_\pi + \rho_{\pi, \tilde{r}} \sigma_{\tilde{r}}}{\sigma_\omega} > 0$. The marginal cost curve is:

$$MC(p) = E(\pi | \omega = p) = \mu_\pi + \rho \frac{\sigma_\pi}{\sigma_\omega} (p - \mu_\omega), \quad (13)$$

which is increasing in p when $\rho_{\pi, \tilde{r}}$ (and thus ρ) is positive. Average costs are

$$AC(p) = E(\pi | \omega \geq p) = \mu_\pi + \rho \sigma_\pi \frac{\phi(\frac{p - \mu_\omega}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega}{\sigma_\omega})}, \quad (14)$$

an endogenously truncated normal distribution and increase in p for positive $\rho_{\pi, \tilde{r}}$.

³⁰As dependence is measured in ranks, the cross partial derivative captures more than the correlation between π and ω : it encodes how jointly heavy-tailed they are.

For parametric copula families these conditions restrict the parameter space, for example requiring the correlation coefficient is positive for the normal copula. This relates to the intuitive misconception that positive correlation is sufficient for adverse selection, however, which in fact a consequence of the conditions in Proposition 4. Furthermore, Appendix D shows how the joint distribution of ω and π can be derived from assumptions on the joint distribution of π and $\tilde{r}$.

4.2. Differences between Mature and Immature Markets

I use superscripts MM and IM to denote the mature and immature markets and the relation $X \preceq Y$ to denote first order stochastic dominance. The key assumption in Section 2.2 is that willingness-to-pay is depressed in immature markets relative to mature markets – lowering demand. This can come from objective differences in risk premia, r , or differences in demand frictions, ε :

ASSUMPTION 1. *Willingness-to-pay is on average larger and demand expands in mature markets: $\omega^{MM} \succeq \omega^{IM}$*

This formalizes how shifts in risk premia and demand frictions affect willingness-to-pay. Instead of demand frictions shifting systematically, consumers may have noisier beliefs about new products, reflected in greater dispersion of demand frictions capturing unbiased but diffuse priors. Assumption 1 is weaker than requiring stochastic dominance of objective risk premia, r , or demand frictions, ε . It only needs to be satisfied *on net* and not offset by changes in sorting.

Unlike Spinnewijn (2017), this doesn't require consumers are unbiased on average. Systematic optimism or pessimism in mature markets is possible, consistent with evidence of choice frictions (Brown et al. 2021), biased beliefs (O'Dea and Sturrock 2023), probability distortions (Barseghyan et al. 2013), and limited consideration (Barseghyan et al. 2021).

ASSUMPTION 2. *Risk types, π_i , have identical marginal distributions in mature and immature markets.*

These assumptions restrict how marginal distributions differ between mature and immature markets. Yet, they do not capture how the sorting of consumers may change as markets mature. Changes in sorting may occur because of changes in consumers' self insurance behaviour or behavioural frictions. This changes how purchasing insurance signals an individual's cost:

ASSUMPTION 3. *Resorting is such that copulas C , describing dependence between π_i and ω_i , satisfy the following concordance ordering (Yanagimoto and Okamoto 1969; Tchen 1980; Scarsini 1984): $C^{IM}(u, v) \geq C^{MM}(u, v) \forall u, v \in [0, 1]$*

This orders joint distributions *between* markets by implied dependence. Costs and willingness-to-pay have larger concordance—their joint density is more concentrated

around a monotone graph.³¹ Sorting on cost has increased, changing how insurance purchase signals cost to insurers.

For transparency, I provide closed form proofs under joint normality (relegating general proofs to Appendix E), though general results rely only on the above assumptions:³²

ASSUMPTION 4 (Joint Normality). π_i and $\tilde{r}_i$ are jointly normally distributed. Additionally, the variance of willingness-to-pay is constant, $\sigma_\omega^{IM} = \sigma_\omega^{MM}$, and correlation between risk type and willingness-to-pay is weaker in mature markets, $\text{corr}(\pi, \omega^{IM}) \geq \text{corr}(\pi, \omega^{MM}) > 0$.

4.3. Equilibrium in Mature and Immature Markets

I now turn to understanding the implication of Assumptions (1) - (3) on quantities, costs, and the market equilibrium.

At any price, average willingness-to-pay is larger in mature markets. It is immediate that more individuals purchase insurance – a *level effect* on demand: $D^{IM}(p) \leq D^{MM}(p) \quad \forall p$.

This level effect does not change the identity of marginal (and inframarginal) consumers. Appendix E derives additional results for $\tilde{r}$ and π directly. As with the discussion in Section 2.2, I now turn to the role of sorting.

PROPOSITION 5 (Increased Adverse Selection). *Consumers are adversely selected in both mature and immature markets. If resorting between markets satisfies the following additional assumption:*

- i. [Monotone Regression Dependence Ordering] $\tilde{g}_u^{IM}(g_u^{MM}(v))$ is increasing in v for all $u \in (0, 1)$, where $g_x(y) = \partial C(x, y) / \partial y$ and $\tilde{g}_x \equiv g_x^{-1}$,

Then consumers are more adversely selected in immature markets $\left(\frac{\partial MC^{IM}(p)}{\partial p} \geq \frac{\partial MC^{MM}(p)}{\partial p} > 0 \right)$

³¹Anderson and Smith (2024) and Boerma et al. (2023) study concordance orderings from assignment problems. Anderson and Smith (2024) shows concordance order increases as local output function synergies increase. Boerma et al. (2023) shows optimal assignments are larger in concordance when mismatch costs become less concave. I show in Appendix E how this can be constructed from assumptions on the dependence between π_i and $\tilde{r}_i$ directly.

³²To simplify exposition Assumption 4 rules out changes in the dispersion of willingness-to-pay across mature and immature markets. Under this condition, changes to the primitives take the form of location shifts for the marginal distributions. This is weaker than the stochastic ordering of normals as it does not impose a common variance across mature and immature markets for π_i and $\tilde{r}_i$.

Constant dispersion in willingness-to-pay is required for the normal distribution as the support is unbounded. Without this, equivalent results are available with truncated tails (Levy 1982).

PROOF. Consider the joint (normal) distribution of cost type and willingness-to-pay,

$$\begin{pmatrix} \pi_i \\ \omega_i \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_\pi \\ \mu_\omega \end{pmatrix}, \begin{pmatrix} \sigma_\pi^2 & \rho \sigma_\pi \sigma_\omega \\ \rho \sigma_\pi \sigma_\omega & \sigma_\omega^2 \end{pmatrix} \right],$$

where

$$\rho = \frac{\sigma_\pi^2 + \rho_{\pi,r} \sigma_\pi \sigma_r}{\sigma_\omega \sigma_\pi} > 0. \quad (15)$$

Then the marginal cost curve has the following closed form expression:

$$MC(p) = E(\pi | \omega = p) = \mu_\pi + \rho \frac{\sigma_\pi}{\sigma_\omega} (p - \mu_\omega), \quad (16)$$

which is increasing in p when ρ is positive. The second part of the proposition then follows from Assumption 4. $\square$

Unfortunately, it is difficult to express the additional assumption required in the general case as lower level restrictions on the joint distribution between cost type and risk premia. While this assumption lacks intuitive appeal, however, it is only required for this specific result because the usual definition of adverse selection invokes the slope of the marginal cost curve. While this is useful to distinguish adverse and advantageous selection, the notion of being *more* selected it gives rise to is too stringent. Instead, the characterisation of the equilibrium in this paper use results on how adverse selection determines average and marginal costs *in levels* and for a *given demand* which are derived under milder assumptions:

LEMMA 1 (Adverse Selection In Levels). *When there is no re-sorting ($C = C^{IM} = C^{MM}$) marginal costs are higher at any price. However, when re-sorting occurs, effects on marginal cost are ambiguous.*

While the slope of the marginal cost curve increases, the effect of re-sorting has an ambiguous effect on the level of marginal costs depending on the price:³³

COROLLARY 2 (Decomposing Marginal Costs). *Under Assumption 4, the differences in marginal costs between mature and immature markets can be decomposed into the level effect*

³³It is always possible to define a cost curve holding sorting fixed, $MC^{MM,nosort}$, and decompose the general case as $\underbrace{(MC^{IM}(p) - MC^{MM,nosort}(p))}_{\text{level effect} \geq 0} + \underbrace{(MC^{MM,nosort}(p) - MC^{MM}(p))}_{\text{re-sorting effect}}$.

of demand and a re-sorting effect

$$MC^{IM}(p) - MC^{MM}(p) = \frac{\sigma_\pi}{\sigma_\omega} \rho^{IM} \left[\underbrace{(\mu_\omega^{MM} - \mu_\omega^{IM})}_{\text{level effect} \geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}}\right) (p - \mu_\omega^{MM})}_{\text{re-sorting effect}} \right]. \quad (17)$$

Absent re-sorting, the increase in marginal costs depends only on the contraction in demand as the second term is equal to zero. However, when there is re-sorting the sign of the second term is ambiguous. For prices above the average willingness-to-pay in mature markets, re-sorting increases the difference in marginal costs between mature and immature markets. When prices are lower, however, the effect is negative as those purchasing insurance at high prices in immature markets are more adversely selected. Consequently, the pool of remaining consumers at low prices have lower costs.

Nevertheless, despite ambiguity at the margin, when consumers are reordered and selection on cost increases, then insurers face higher *average* costs:

PROPOSITION 6 (Cost Expansion). *Average costs are increasing in price and early adopters are higher cost on average, $AC^{IM}(p) \geq AC^{MM}(p)$.*

PROOF. Under joint normality, average costs are the expected value of a truncated normal distribution. The average cost curve is

$$AC(p) = E(\pi | \omega \geq p) = \mu_\pi + \rho \sigma_\pi \frac{\phi(\frac{p - \mu_\omega}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega}{\sigma_\omega})}. \quad (18)$$

The average cost depends on the inverse Mills ratio of the subjective willingness-to-pay, which captures the extent to which selection truncates the distribution of cost types.

Comparing across markets gives $\forall p$

$$\frac{AC^{IM}(p) - AC^{MM}(p)}{\sigma_\pi \rho^{IM}} = \underbrace{\left(\frac{\phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})} - \frac{\phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})} \right)}_{\text{level effect} \geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}} \right) \frac{\phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})}}_{\text{re-sorting effect} \geq 0} \geq 0 \quad (19)$$

The inverse Mills ratio and the difference in the inverse Mills ratio are always positive by the log concavity of the normal distribution ensuring non-negativity. $\square$

In contrast to marginal costs, both effects work in the same direction (raising average

costs at a given price), as additional selection on costs among inframarginal consumers dominates the ambiguous effect on the marginal consumer. The size of both effects are increasing in the share of heterogeneity in willingness-to-pay coming from cost type, π , while resorting effects depend on the amount of selection in immature markets. When this selection is large, insurance purchase is a stronger signal of cost. Consequently, increased sorting magnifies existing selection.

The key insight is that increased sorting on cost in immature markets makes insurance purchase signal higher costs. As sorting increases, consumers with highest willingness-to-pay have highest costs, leading to higher marginal costs at low quantities:

PROPOSITION 7 (Cost Expansion Fixing Demand). *Average Costs as a function of quantity are larger in immature markets, $AC^{IM}(q) \geq AC^{MM}(q)$. Marginal Costs rotate around a point and are higher at low quantities, but smaller at high quantities, thus $MC^{IM}(q) \geq MC^{MM}(q)$ for all $q \leq \tilde{q}$ and vice versa. Absent re-sorting, both cost curves are identical.*

Moreover, the adverse selection wedge is higher (and more negative) at any quantity demanded: $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$.

PROOF. Substituting inverse demand into (18) gives average costs as a function of the quantity demanded,

$$AC(q) \equiv E\left(\pi | \omega \geq D^{-1}(q)\right) = \mu_\pi + \rho\sigma_\pi \frac{\phi(\Phi^{-1}(1-q))}{1 - \Phi(\Phi^{-1}(1-q))} = \mu_\pi + \rho\sigma_\pi \frac{\phi(\Phi^{-1}(1-q))}{q}. \quad (20)$$

Differences in average costs are then

$$AC^{IM}(q) - AC^{MM}(q) = \underbrace{\rho^{IM}\sigma_\pi \left(1 - \frac{\rho^{MM}}{\rho^{IM}}\right) \frac{\phi(\Phi^{-1}(1-q))}{q}}_{\text{re-sorting effect}} \geq 0 \quad \forall q. \quad (21)$$

Equivalently, substituting inverse demand into (13) gives marginal costs as a function of quantity,

$$MC(q) \equiv E\left(\pi | \omega = D^{-1}(q)\right) = \mu_\pi + \rho\sigma_\pi \Phi^{-1}(1-q). \quad (22)$$

It follows that

$$MC^{IM}(q) - MC^{MM}(q) = \underbrace{(\rho^{IM} - \rho^{MM})(\sigma_\pi \Phi^{-1}(1-q))}_{\text{re-sorting effect}}$$

which shows that the marginal costs are higher at any quantity demanded below the median. This is because the final term is the quantile function of the standard normal distribution (evaluated at $1 - q$), thus it is negative when $1 - q < 1/2$. This corresponds to the rotation of marginal costs around the median. $\square$

These results establish the prerequisites for Section 2.2, so remaining results follow directly for the general case.

4.4. Differentiated Goods and Market Power

I now extend the model by relaxing product homogeneity and allow for market power. There are $j \in \{1, \dots, J\}$ firms offering identical insurance contracts bundled with firm specific differentiated attributes X_j . As in the equilibrium described above, firms maximize profits by setting a uniform price. Under Bertrand-Nash price competition, firm j 's equilibrium pricing equation (replacing equation 6) becomes:

$$p_j^* - AC(p_j^*) = \frac{1}{\eta_{D,p_j}} - \left(AC(p_j^*) - MC(p_j^*) \right), \quad (23)$$

where the second term captures pricing pressure from adverse selection. Rearranging yields the standard monopolist pricing:

$$p_j^* = \frac{1}{\eta_{D,p_j}} + MC(p_j^*). \quad (24)$$

As in Mahoney and Weyl (2017), the impact of increased adverse selection depends on the quantities in the mature market:

LEMMA 2 (Under-Insurance with Product Differentiation). *Assume quantities in the mature market are small ($q_j^{MM} < \tilde{q}$) and elasticity advantages are modest ($\eta_{D,p_j}^{MM}/\eta_{D,p_j}^{IM} \geq 1 - \eta_{D,p_j}^{MM} [MC^{MM}(q_j) - MC^{IM}(q_j)]$ for $q_j < \tilde{q}$).*

It follows that, if an equilibrium exists, equilibrium prices are higher and quantities lower in immature markets: $q_j^{MM} \geq q_j^{IM}$ and $p_j^{MM} \leq p_j^{IM}$.

This confirms that adverse selection reduces firms' incentive to supply differentiated products in immature markets, generating higher prices, lower quantities, reduced profits, and lower consumer surplus.³⁴

³⁴Corollary A2 in Appendix E provides conditions guaranteeing this. These are satisfied under constant shape restrictions, e.g. stochastic dominance for log-concave willingness-to-pay distributions (Milgrom and Weber 1982; Chade and Schlee 2012; Chade and Swinkels 2021).

5. Discussion

This paper shows the existence of a novel dynamic free-rider problem that can arise in adversely selected markets. When new products launch, temporarily low demand or stronger sorting increases adverse selection among early adopters. Initial entrants bear these losses, while later competitors bid away mature-market rents. Because long-run rents cannot finance early losses, cross-subsidization fails and entry can unravel—even when the mature market would be efficient. Counter-intuitively, market power can improve outcomes by enabling intertemporal cross-subsidies across cohorts.

I formalize the mechanism in a simple entry model and microfound period profits from equilibrium prices and quantities in a standard adverse-selection setting with multidimensional heterogeneity. Under economically motivated assumptions on risk premia and potential behavioral biases, early adopters impose higher costs despite an unchanged risk distribution, amplifying under-insurance and depressing welfare in immature markets. Comparative statics cover monotonicity with multi-dimensional heterogeneity, yielding a partial ordering of environments—and, hence, of equilibrium prices, quantities, selection, and welfare—that maps cleanly into the the popular graphical framework for insurance analysis (Einav and Finkelstein 2011).

These results microfound “missing markets” extending classic unraveling to a dynamic setting and jointly rationalizing low utilization, slow take-up, and market inactivity. An application to the emerging market for deferred income annuities shows that this is an empirically relevant concern. Selection declines as the market deepens; using data on market size, contract terms, and insurer costs, I distinguish this pattern from alternative price dynamics. Although the paper focuses on insurance, the logic applies to selection markets more broadly—whenever buyers impose heterogeneous costs on sellers—regardless of whether immaturity distorts true welfare or only perceived value. The general framework also delivers predictions across time and place or other variations in selection markets; for example, shifts in risk premia reduce adverse selection and sharpen implications for the lemons penalty as in Blundell et al. (2019).

References

Aghion, P., A. Bergeaud, and J. Van Reenen (2023). The impact of regulation on innovation. *American Economic Review* 113(11), 2894–2936.

- Akerlof, G. A. (1970). The market for "lemons": Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics* 84(3), 488–500.
- Ameriks, J., J. Briggs, A. Caplin, M. D. Shapiro, and C. Tonetti (2016). The long-term-care insurance puzzle: Modeling and measurement. Technical report, National Bureau of Economic Research.
- Anderson, A. and L. Smith (2024). The comparative statics of sorting. *American Economic Review* 114(3), 709–751.
- Attar, A., T. Mariotti, and F. Salanié (2021). Entry-proofness and discriminatory pricing under adverse selection. *American Economic Review* 111(8), 2623–2659.
- Azevedo, E. M. and D. Gottlieb (2017). Perfect competition in markets with adverse selection. *Econometrica* 85(1), 67–105.
- Barseghyan, L., F. Molinari, T. O’Donoghue, and J. C. Teitelbaum (2013). The nature of risk preferences: Evidence from insurance choices. *American economic review* 103(6), 2499–2529.
- Barseghyan, L., F. Molinari, and M. Thirkettle (2021). Discrete choice under risk with limited consideration. *American Economic Review* 111(6), 1972–2006.
- Benhabib, J., J. Perla, and C. Tonetti (2021). Reconciling models of diffusion and innovation: A theory of the productivity distribution and technology frontier. *Econometrica* 89(5), 2261–2301.
- Blundell, R., R. Gu, S. Leth-Petersen, H. Low, and C. Meghir (2019). Durables and lemons: private information and the market for cars. Technical report, National Bureau of Economic Research.
- Boerma, J., A. Tsyvinski, R. Wang, and Z. Zhang (2023). Composite sorting. Technical report, National Bureau of Economic Research.
- Boyer, M. M., P. De Donder, C. Fluet, M.-L. Leroux, and P.-C. Michaud (2020). Long-term care insurance: Information frictions and selection. *American Economic Journal: Economic Policy* 12(3), 134–169.
- Briggs, J., C. Rogers, and C. Tonetti (2023). Risky insurance: Life-cycle insurance portfolio choice with incomplete markets.
- Brown, J. R., A. Kapteyn, E. F. P. Luttmer, O. S. Mitchell, and A. Samek (2021). Behavioral Impediments to Valuing Annuities: Complexity and Choice Bracketing. *The Review of Economics and Statistics* 103(3), 533–546.
- Brown, J. R. and J. M. Poterba (2006). Household ownership of variable annuities. *Tax Policy and the Economy* 20, 163–191.
- Cannon, E. and I. Tonks (2016). Cohort mortality risk or adverse selection in annuity markets? *Journal of Public Economics* 141, 68–81.
- Chade, H. and E. Schlee (2012). Optimal insurance with adverse selection. *Theoretical Economics* 7(3), 571–607.
- Chade, H. and J. Swinkels (2021). Screening in vertical oligopolies. *Econometrica* 89(3), 1265–1311.
- Chetty, R., J. N. Friedman, and E. Saez (2013). Using differences in knowledge across neighborhoods to uncover the impacts of the eitc on earnings. *American Economic Review* 103(7), 2683–2721.
- Cocco, J. F. and P. Lopes (2015). Reverse mortgage design. In *2015 Meeting Papers*, Volume 632. Citeseer.

- Cocco, J. F. and P. Lopes (2020). Aging in place, housing maintenance, and reverse mortgages. *The Review of Economic Studies* 87(4), 1799–1836.
- Cohen, A. and L. Einav (2007). Estimating risk preferences from deductible choice. *American Economic Review* 97(3), 745–788.
- Dorfman, M. S. (1971). The product performance of the life insurance industry: Revisited. *Journal of Risk and Insurance*, 613–626.
- Duflo, E. and E. Saez (2003). The role of information and social interactions in retirement plan decisions: Evidence from a randomized experiment. *The Quarterly Journal of Economics* 118(3), 815–842.
- Egan, M., S. Ge, and J. Tang (2022). Conflicting interests and the effect of fiduciary duty: Evidence from variable annuities. *The Review of Financial Studies* 35(12), 5334–5386.
- Einav, L. and A. Finkelstein (2011). Selection in insurance markets: Theory and empirics in pictures. *Journal of Economic perspectives* 25(1), 115–138.
- Einav, L. and A. Finkelstein (2023). Empirical analyses of selection and welfare in insurance markets: a self-indulgent survey. *The Geneva Risk and Insurance Review* 48(2), 167–191.
- Einav, L., A. Finkelstein, and M. R. Cullen (2010). Estimating welfare in insurance markets using variation in prices. *The Quarterly Journal of Economics* 125(3), 877–921.
- Einav, L., A. Finkelstein, and N. Mahoney (2021). The IO of selection markets. In *Handbook of Industrial Organization*, Volume 5, pp. 389–426. Elsevier.
- Einav, L., A. Finkelstein, and P. Schrimpf (2010). Optimal mandates and the welfare cost of asymmetric information: Evidence from the uk annuity market. *Econometrica* 78(3), 1031–1092.
- Fang, H. and Z. Wu (2018). Multidimensional private information, market structure, and insurance markets. *The RAND Journal of Economics* 49(3), 751–787.
- Finkelstein, A. and J. Poterba (2002). Selection effects in the united kingdom individual annuities market. *The Economic Journal* 112(476), 28–50.
- Finkelstein, A. and J. Poterba (2004). Adverse selection in insurance markets: Policyholder evidence from the uk annuity market. *Journal of Political Economy* 112(1), 183–208.
- Geruso, M. and T. J. Layton (2017). Selection in health insurance markets and its policy remedies. *Journal of Economic Perspectives* 31(4), 23–50.
- Geruso, M., T. J. Layton, G. McCormack, and M. Shepard (2023). The two-margin problem in insurance markets. *The Review of Economics and Statistics* 105(2), 237–257.
- Gola, P. (2021). Supply and demand in a two-sector matching model. *Journal of Political Economy* 129(3), 940–978.
- Griffith, R. and G. Macartney (2014). Employment protection legislation, multinational firms, and innovation. *Review of Economics and Statistics* 96(1), 135–150.
- Handel, B., I. Hendel, and M. D. Whinston (2015). Equilibria in health exchanges: Adverse selection versus reclassification risk. *Econometrica* 83(4), 1261–1313.
- Handel, B. R. (2013). Adverse selection and inertia in health insurance markets: When nudging hurts. *American Economic Review* 103(7), 2643–2682.
- Handel, B. R., J. T. Kolstad, and J. Spinnewijn (2019). Information frictions and adverse selection: Policy interventions in health insurance markets. *Review of Economics and Statistics* 101(2),

326–340.

- Hendren, N. (2013). Private information and insurance rejections. *Econometrica* 81(5), 1713–1762.
- Hosseini, R., K. A. Kopecky, and K. Zhao (2022). The evolution of health over the life cycle. *Review of Economic Dynamics* 45, 237–263.
- Israel, M. (2005). Services as experience goods: An empirical examination of consumer learning in automobile insurance. *American Economic Review* 95(5), 1444–1463.
- Kim, D. (2021). Partially identifying competing risks models: An application to the war on cancer. *Journal of Econometrics*.
- Koijen, R. S. and M. Yogo (2015). The cost of financial frictions for life insurers. *American Economic Review* 105(1), 445–475.
- Kong, E., T. Layton, and M. Shepard (2024). Adverse selection and (un) natural monopoly in insurance markets. Technical report, National Bureau of Economic Research.
- Leive, A., L. Friedberg, and B. Davis (2022). Overpaying and undersaving? correlated mistakes in retirement saving and health insurance choices. Technical Report 2022-06, Wharton Pension Research Council Working Paper.
- Levy, H. (1982). Stochastic dominance rules for truncated normal distributions: A note. *The Journal of Finance* 37(5), 1299–1303.
- Lockwood, L. M. (2012). Bequest motives and the annuity puzzle. *Review of Economic Dynamics* 15(2), 226–243.
- Lockwood, L. M. (2018). Incidental bequests and the choice to self-insure late-life risks. *American Economic Review* 108(9), 2513–2550.
- Mahoney, N. and E. G. Weyl (2017). Imperfect competition in selection markets. *Review of Economics and Statistics* 99(4), 637–651.
- Michaud, P.-C. and P. S. Amour (2023). Longevity, health and housing risks management in retirement. Technical report, National Bureau of Economic Research.
- Milgrom, P. R. (1981). Good news and bad news: Representation theorems and applications. *The Bell Journal of Economics*, 380–391.
- Milgrom, P. R. and R. J. Weber (1982). A theory of auctions and competitive bidding. *Econometrica*, 1089–1122.
- Mitchell, O. S., J. M. Poterba, M. J. Warshawsky, and J. R. Brown (1999). New evidence on the money's worth of individual annuities. *American Economic Review* 89(5), 1299–1318.
- Murray, M. L. (1976). The theory and practice of innovation in the private insurance industry. *Journal of Risk and Insurance*, 653–671.
- Nakajima, M. and I. A. Telyukova (2017). Reverse mortgage loans: A quantitative analysis. *The Journal of Finance* 72(2), 911–950.
- O'Dea, C. and D. Sturrock (2023). Survival pessimism and the demand for annuities. *The Review of Economics and Statistics* 105(2), 442–457.
- Pashchenko, S. (2013). Accounting for non-annuitization. *Journal of Public Economics* 98, 53–67.
- Poterba, J. M. and A. Solomon (2021). Discount rates, mortality projections, and money's worth calculations for us individual annuities. Technical report, National Bureau of Economic Research.
- Rothschild, M. and J. Stiglitz (1976). Equilibrium in competitive insurance markets: An essay on

- the economics of imperfect information. *The Quarterly Journal of Economics* 90(4), 629–649.
- Rudelius, W. and G. L. Wood (1970). Life insurance product innovations. *Journal of Risk and Insurance*, 169–184.
- Russo, N., R. McGee, M. De Nardi, M. Borella, and R. Abram (2024). Health inequality and economic disparities by race, ethnicity, and gender. Technical report, National Bureau of Economic Research.
- Scarsini, M. (1984). On measures of concordance. *Stochastica* 8(3), 201–218.
- Solomon, A. (2023). Imperfect private information in insurance markets. *Review of Economics and Statistics*, 1–46.
- Spinnewijn, J. (2017). Heterogeneity, demand for insurance, and adverse selection. *American Economic Journal: Economic Policy* 9(1), 308–343.
- Starc, A. (2014). Insurer pricing and consumer welfare: Evidence from medigap. *The RAND Journal of Economics* 45(1), 198–220.
- Tchen, A. H. (1980). Inequalities for distributions with given marginals. *The Annals of Probability*, 814–827.
- Thompson, J. R. (2010). Counterparty risk in financial contracts: Should the insured worry about the insurer? *The Quarterly Journal of Economics* 125(3), 1195–1252.
- Veiga, A. and E. G. Weyl (2016). Product design in selection markets. *The Quarterly Journal of Economics* 131(2), 1007–1056.
- Warshawsky, M. (1988). Private annuity markets in the united states: 1919-1984. *Journal of Risk and Insurance*, 518–528.
- Webb, A. (2011). The United States Longevity Insurance Market. In *Securing Lifelong Retirement Income: Global Annuity Markets and Policy*. Oxford University Press.
- Williams, H. L. (2017). How do patents affect research investments? *Annual Review of Economics* 9, 441–469.
- Yanagimoto, T. and M. Okamoto (1969). Partial orderings of permutations and monotonicity of a rank correlation statistic. *Annals of the Institute of Statistical Mathematics* 21, 489–506.
- Zinman, J. (2014). Consumer credit: Too much or too little (or just right)? *The Journal of Legal Studies* 43(S2), S209–S237.

Online Appendix

Appendix A. Proofs for Section 1

Proof of the first part. In immature markets let $-C < 0$ denote the loss when both firms enter and $\Pi_M^{IM} < 0$ denote the monopolist profit, respectively. Equivalently Π_M^{MM} denotes the monopolist profit and $\Pi^{MM} = 0$ denotes the (competitive) profit in mature markets.

	enter	exit		enter	exit
enter	$(0, 0)$	$(\Pi_M^{MM}, 0)$	enter	$(-C, -C)$	$(\Pi_M^{IM}, 0)$
exit	$(0, \Pi_M^{MM})$	$(0, 0)$	exit	$(0, \Pi_M^{IM})$	$(0, 0)$
(a) Mature Market			(b) Immature Market		

TABLE A1. Second Period Payoff Matrix

Payoffs in the second subgame when at least one firm entered in the first period are given in Panel A of Table A1.

In a mature market that does not fully unravel $\Pi_M^{MM} > 0$ and entering is a profitable deviation for both firms if neither enter, thus this cannot be a nash equilibrium of the subgame. There is no profitable deviation from the symmetric strategy profile (enter, enter) as each firm also earns zero under unilateral deviation.

Next, consider the second subgame when neither firm entered in the first period (Panel B). When $\Pi_M^{IM} < 0$, exiting is a strictly dominant strategy and there is a unique equilibrium.

Taking these contingent strategy profiles as given, the normal form representation of the first subgame is in Table A2. As the market is immature and firms earn zero equilibrium profits in mature markets, these payoffs are identical to the immature market second period subgame and exit is strictly dominant.

	enter	exit
enter	$(-C + 0, -C + 0)$	$(\Pi_M^{IM} + 0, 0 + 0)$
exit	$(0 + 0, \Pi_M^{IM} + 0)$	$(0 + 0, 0 + 0)$

TABLE A2. First Period

Therefore, the following symmetric strategy profile is an SPE:

- First period: Play *exit*
- Second period: Play *exit* if no firm has entered, otherwise *enter*

and in equilibrium no firm enters in either period.

Proof of second part. When the market is mature in the second period, the SPE has firms playing strictly dominant strategies. Therefore, only refinements for the immature second period subgame need to be considered.

As this is a two player, two action game, the set of trembling hand perfect equilibria is equivalent to the set of equilibria in which no player plays a weakly dominated strategy. The equilibria considered above (exit, exit) satisfies this. However, the asymmetric strategy profiles in the second round, either (entry, exit) or (exit, entry), do not because entering is weakly dominated by exiting.

This leaves a unique equilibria surviving the trembling-hand refinement in this subgame. The same argument can be applied in the same fashion to the first period delivering a unique equilibria of the extensive form game.

Appendix B. Proofs & Additional Results for Section 2.2

PROOF OF PROPOSITION 2. Let p^{MM} be the equilibrium price in the mature market,

$$D^{MM}(p^{MM}) = AC^{MM}(p^{MM}) . \quad (A1)$$

If it is also an equilibrium price in the immature market, then

$$D^{IM}(p^{MM}) = AC^{IM}(p^{MM}) \quad (A2)$$

also holds. But the demand contraction and increased sorting¹ imply the following inequality must also hold

$$D^{IM}(p^{MM}) \leq D^{MM}(p^{MM}) = AC^{MM}(p^{MM}) \leq AC^{IM}(p^{MM}) , \quad (A3)$$

which is a contradiction unless the immature and mature markets are identical.

Furthermore, if there exists a price $p^{IM} \neq p^{MM}$ that clears the immature market then differences across markets satisfy *strict* dominance. Therefore

$$D^{IM}(p^{IM}) = AC^{IM}(p^{IM}) \longleftrightarrow p^{IM} > p^{MM} , \quad (A4)$$

and

$$q^{IM} = D^{IM}(p^{IM}) < D^{MM}(p^{IM}) < D^{MM}(p^{MM}) = q^{MM} . \quad (A5)$$

□

PROOF OF PROPOSITION 3. First, note that firms always earn zero profits in equilibrium. Consequently, consumer surplus is sufficient to pareto order equilibria. For a demand level q^* , consumer surplus in immature markets is defined as

$$CS^{IM}(q^*) = \int_0^{q^*} \tilde{D}^{IM}(q) - MC^{IM}(q) dq \quad \text{where} \quad \tilde{D}(q) = D^{-1}(q), \quad (A6)$$

under the revealed preference benchmark. It follows that the difference in consumer surplus due to increased adverse selection is

¹This will be formalised in the general case as Propositions A1 and 6.

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_0^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq - \int_0^{q^{IM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq, \quad (A7)$$

and given $q^{MM} \geq q^{IM}$ (Proposition 2) this can be expressed as

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq \geq 0, \quad (A8)$$

which captures the expanded coverage of insurance in mature markets. Figure 3 Panel A illustrates this graphically where the red shaded area to the right of q^* denotes the foregone surplus due to the additional under-insurance in the short run – the area between the marginal cost curve and the immature market demand curve.

For the paternalistic benchmark which uses the mature market inverse demand

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_{q^{IM}}^{q^{MM}} \tilde{D}^{MM}(q) - MC^{IM}(q) dq = \quad (A9)$$

$$\underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{MM}(q) - \tilde{D}^{IM}(q) dq}_{\text{Paternalism } (\geq 0)} + \underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq}_{\text{Case (i)}} \geq 0, \quad (A10)$$

where the positive paternalistic contribution follows from the contraction of demand (Proposition A1 in the general analysis below). Figure 3 Panel B shows this graphically. $\square$

Appendix C. Additional Discussion of Section 3

Testimonials. To highlight the widespread use of the Annuity Shopper quotes I reproduced the following testimonial from the July 2005 issue. The testimonial is addressed to Hersh Stein (the publisher):

Dear Hersh:

Thank you for providing the WebAnnuties packet and the Annuity Shopper.

It is the best source of information I have found on the subject of annuities. And a special thanks for calling and answering my questions.

As an attorney who frequently represents clients regarding investments, retirement and estate planning I will highly recommend you for their annuity needs. As I mentioned I have several annuities now and will continue to add more to my own retirement portfolio.

I have recommended you and your company to other estate planners and financial advisors.

Very truly yours, David D. Dunakey

Similar testimonies are frequently published.

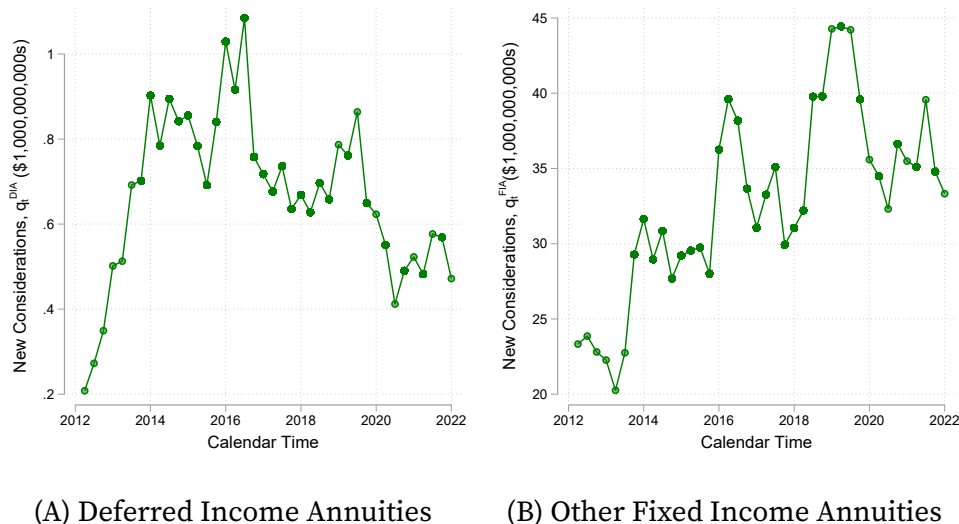


FIGURE A1. Growth In The Deferred Income Annuities Market

Sales of Annuity Products. Figure A1 shows how the size of this market has evolved over time with the left panel displaying new deferred income annuity sales. At the start of this period the market for deferred income annuities was small, totaling around \$200 million and accounting for less than 1% of the overall fixed income annuity market. Subsequently, however, the market expanded rapidly between 2012 and 2014, with the real value of new considerations more than quadrupling in size. The right panel shows sales of all other fixed income annuity products, the most similar alternative product in the annuity space. Comparing the two panels suggests an element of common cyclical, but growth in fixed income annuities is much smaller over the period.

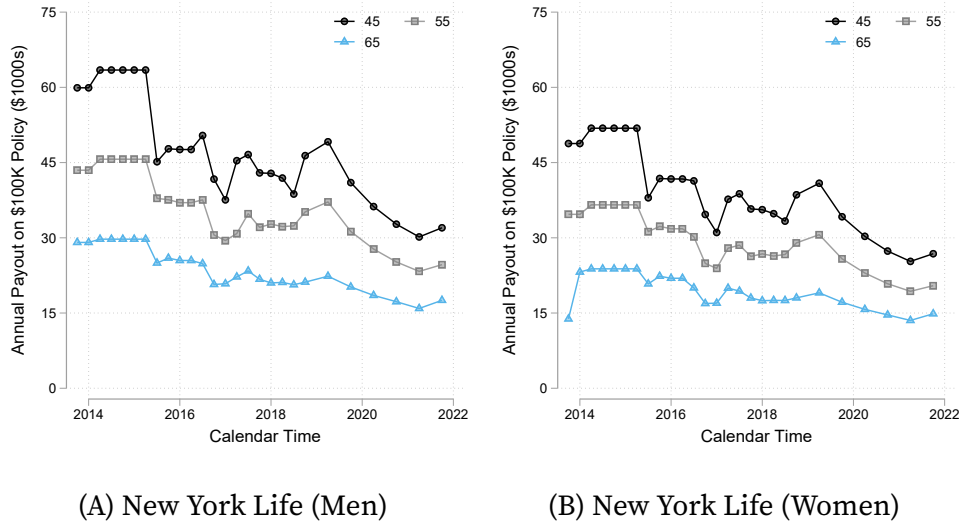


FIGURE A2. Average Payouts from Deferred Income Annuity Policies Beginning Payment at Age 80 By Gender and Purchase Age

Within Insurer Payout Variation. Figure A2 repeats the exercise in Figure 5 for contracts offered by New York Life, but conditioning on the deferral age and varying the purchase age. The left panel shows contracts for men and the right panel for women with each line corresponding to a different purchase age between 45 and 65. Again, contracts that offer payouts further into the future have higher payout amounts. Similarly, payouts for men who have lower expected longevity at all ages are smaller than for women. Despite systematic differences by product characteristics, a notable feature of both Figures 5 and A2 is evidence of a market level trend in payouts over time for all contract terms and for both insurers. This reflects that the largest cost to the insurer is the opportunity cost of foregone investment (see Mitchell et al. 1999) as well as a changing pool of annuitants.

C.1. Additional Empirical Results for Section 3

Table A3 presents alternative specifications corresponding to the results in Table 1. The first column repeats the baseline estimate from the final specification with the full set of controls in Table 1. The second column allows for a non-parametric time trend by replacing the control for contemporaneous demand in all other fixed income annuity markets with a series of quarterly indicators. The parameter estimate is almost unchanged, highlighting that identification does not rely on the parametric control for aggregate market conditions.

	(1) Base	(2) Time FEs	(3) Levels	(4) IAM Life Table	(5) Policy FEs
$\mathbb{1}[a > 45] \ln Q_t$	-0.238*** (0.027)	-0.245*** (0.024)	-0.046*** (0.005)	-0.064*** (0.008)	-0.234*** (0.027)
N	5, 857	5, 857	5, 882	5, 882	5, 857
R ²	0.514	0.653	0.666	0.395	0.522

Notes: * $p < 0.10$, ** $p < 0.5$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient δ in equation (10). Implied selection is recovered using equation (9) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and aggregate demand for all other FIA products (excluding (2) due to colinearity with the time specific fixed effects).

TABLE A3. Adverse Selection and Market Maturity

The third column replaces the dependent variable with selection measured as deviations in levels rather than logs. The magnitude of the estimate changes due to the alternative specification of the dependent variable, but the qualitative patterns remain. The fourth column uses life tables from the individual Annuitant Mortality (IAM) 2012 table, which is described in American Academy of Actuaries / Society of Actuaries (2011). As discussed by [Poterba and Solomon \(2021\)](#) this forecasts significant mortality improvement at odds with observed mortality improvements. Consequently, the overall money's worth measure of annuities is more positive and the log specification discards a large share of the observations where the implied NPV exceeds the cost to consumers. Nevertheless, results across columns 3 and 4, both using levels, suggest that there are similar qualitative findings whether deviations are measured from the pool of all potential consumers or the pool of typical annuitants.

Finally, the fifth column replaces the controls for policy characteristics with a finer set of interact-characteristic fixed effects. Each annuity policy is identified by an insurer-gender-deferral age specific identifier and this specification controls for fixed effects along this interacted dimension. This allows for more granular differences in how insurers market to or administer plans by gender and deferral age which may drive

estimates of the effect of maturity due to differential entry and exit of insurers and policy combinations. The point estimate is almost identical to the baseline specification.

Table A4 repeats this exercise for the results reported in Table 2 which allow for heterogeneous effects by purchase age. Once again the results are robust to alternative controls and definitions of the dependent variable. Point estimates for specifications with alternative controls are almost identical and alternative specifications of the dependent variable yield similar qualitative findings that echo results reported in Table 1. Across all specifications in both Tables A3 and A4 the estimated effects remain economically significant and statistically significant at the 1% level.

	(1) Base	(2) Time FEs	(3) Levels	(4) IAM Life Table	(5) Policy FEs
$\mathbb{1}[a = 55] \ln Q_t$	-0.171*** (0.029)	-0.176*** (0.024)	-0.031*** (0.005)	-0.041*** (0.009)	-0.168*** (0.028)
$\mathbb{1}[a = 65] \ln Q_t$	-0.323*** (0.027)	-0.334*** (0.025)	-0.065*** (0.005)	-0.093*** (0.010)	-0.320*** (0.027)
N	5, 857	5, 857	5, 882	5, 882	5, 857
R ²	0.519	0.658	0.669	0.399	0.527

Notes: * $p < 0.10$, ** $p < 0.5$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient δ_{55} and δ_{65} in equation (11). Implied selection is recovered using equation (9) and panel data collected from policy quotes printed in *Annuity Shopper* magazine.

Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and aggregate demand for all other FIA products (excluding (2) due to colinearity with the time specific fixed effects).

TABLE A4. Adverse Selection and Market Maturity by private information

Appendix D. Proofs & Additional Results for Section 4.1

The Role of the MLRP assumption. The MLRP property is widely invoked in studies of incomplete information and has a convenient intuitive interpretation. The proof of Proposition 4 establishes that monotonicity of the cost curves are each available under milder assumptions on the copula that are directly implied by the stronger MLRP.

PROOF OF PROPOSITION 4. Preliminaries. The monotone likelihood ratio dependence implies (see, e.g., Nelsen 2006 Theorem 5.2.19):

- i. [Component-wise Concavity] $\hat{C}(u, v)$ is concave in v for all $u \in (0, 1)$.
- ii. [Corner Set Monotonicity] $(u - \hat{C}(u, 1 - v))/v$ is increasing in v for all $u \in (0, 1)$.

Proof of Part (i). Beginning with the monotonicity of marginal costs – the CDF of the conditional random variable $\pi|\omega = p$ is

$$\begin{aligned} F_{\pi|\omega}(\pi|\omega) &\equiv \int_{-\infty}^{\pi} f_{\pi|\omega}(u|\omega) du = \int_{-\infty}^{\pi} \frac{f_{\pi, \omega}(u, \omega)}{f_{\omega}(\omega)} du \\ &= C_2(u, F_{\omega}(\omega)) = g(F_{\pi}(\pi)), \end{aligned} \quad (\text{A11})$$

for a given copula C where $C_2(x, y) = \partial C(x, y)/\partial y$.

As the marginal cost curve is a conditional expectation $MC(p) = E(\pi|\omega = p)$, stochastic dominance of the conditional distributions is a sufficient condition to order marginal costs in price

$$F_{\pi|\omega}(\pi|\omega = v_1) \succeq F_{\pi|\omega}(\pi|\omega = v_2) \quad \forall v_1 \geq v_2 \quad (\text{A12})$$

$$\longrightarrow MC(v_1) = E(\pi|\omega = v_1) \geq E(\pi|\omega = v_2) = MC(v_2) \quad \forall v_1 \geq v_2. \quad (\text{A13})$$

Substituting equation (A11) gives the following necessary and sufficient condition:

$$C_2(u, F_{\omega}(v_1)) \leq C_2(u, F_{\omega}(v_2)) \quad \forall v_1 \geq v_2 \text{ and } \forall u \quad (\text{A14})$$

$$\longleftrightarrow C_2(u, a) \leq C_2(u, b) \quad \forall a \geq b \text{ and } \forall u \quad (\text{A15})$$

as $v_1 \geq v_2 \longrightarrow F_{\omega}(v_1) \geq F_{\omega}(v_2)$. This is satisfied for all copulas where $C_2(\cdot, \cdot)$ is decreasing in its second argument. Condition (i) of Proposition (4) guarantees C is concave and satisfies this property.

This establishes that $MC(p)$ is increasing in p .

Proof of Part (ii). Turning to the monotonicity of average costs – the CDF of the conditional random variable $\pi|\omega > p$ is

$$F_{\pi|\omega > p}(\pi|\omega > p) \equiv \frac{u - C(u, F_\omega(p))}{1 - F_\omega(p)}, \quad (\text{A16})$$

for a given copula C .

As the average cost curve is defined as a conditional expectation $AC(p) = E(\pi|\omega \geq p)$, stochastic dominance of the conditional distributions is a sufficient condition to order average costs in price

$$F_{\pi|\omega \geq p}(\pi|\omega \geq v_1) \succeq F_{\pi|\omega \geq p}(\pi|\omega \geq v_2) \quad \forall v_1 \geq v_2 \quad (\text{A17})$$

$$\longrightarrow AC(v_1) = E(\pi|\omega \geq v_1) \geq E(\pi|\omega \geq v_2) = AC(v_2) \quad \forall v_1 \geq v_2 \quad (\text{A18})$$

Substituting equation (A16) gives the following necessary and sufficient condition for this stochastic dominance:

$$F_{\pi|\omega \geq p}(\pi|\omega \geq v_1) \leq F_{\pi|\omega \geq p}(\pi|\omega \geq v_2) \quad \forall v_1 \geq v_2 \text{ and } \forall \pi \quad (\text{A19})$$

$$\frac{u - C(u, F_\omega(v_1))}{1 - F_\omega(v_1)} \leq \frac{u - C(u, F_\omega(v_2))}{1 - F_\omega(v_2)} \quad \forall v_1 \geq v_2 \text{ and } \forall u \quad (\text{A20})$$

as $v_1 \geq v_2 \longrightarrow F_\omega(v_1) \geq F_\omega(v_2)$, condition (ii) of Proposition (4) guarantees C satisfies this property, delivering an ordering of average costs for a given copula.

Proof of Part (iii).

Parts (i) and (ii) establish the monotonicity of $MC(p)$ and $AC(p)$. Note that using the law of iterated expectations we can equivalently express the average cost as

$$AC(p) = E(\pi|\omega \geq p) = E(MC(p)|\omega \geq p) \geq MC(p), \quad (\text{A21})$$

where the final inequality follows directly from monotonicity. $\square$

D.1. Additional Results and Discussion of Copula

This section shows how to derive the distribution of willingness-to-pay directly from assumptions on cost and risk premia in the linear case. This will depend on the dependence between cost and risk premia, as encoded by the copula $\tilde{C}$, and marginal distributions. The following results are derived in [Cherubini et al. \(2011\)](#) who also provide extensions to multivariate settings (see also [Navarro and Sarabia 2022](#)). These multivariate extensions can also be used to derive expressions for the joint-density of net premia from the underlying densities of risk premia r and choice frictions ε . This extends general results on the convolution of probability distributions.

The distribution function for the ω is given by F_ω

$$F_\omega(\omega) = \int_{-\infty}^{\infty} f_\pi(c) \partial_1 \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)) dc \quad (\text{A22})$$

provided that $\lim_{v \rightarrow 0^+} \partial_1 \tilde{C}(u, v) = 0$ for all $u \in (0, 1)$. This is a mixture over the conditional distribution of $(\tilde{r} + c | \pi = c)$, which is given by²

$$F_{\omega|\pi}(\omega | \pi = c) = \partial_1 \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)). \quad (\text{A23})$$

We can use this to derive the following joint density function

$$F_{\pi, \omega}(t, v) = \int_{-\infty}^t F_{\omega|\pi}(v | t) dF_\pi(t) = \int_{-\infty}^t \partial_1 \tilde{C}(F_\pi(t), F_{\tilde{r}}(v - t)) dF_\pi(t), \quad (\text{A24})$$

which in turn can be used recover C , the copula specifying the dependence between ω and π , by application of Sklar's Theorem:

$$C(u, v) = \int_{-\infty}^v \partial_1 \tilde{C}(x, F_{\tilde{r}}(F_\omega^{-1}(u) - F_\pi^{-1}(x))) dx. \quad (\text{A25})$$

There are not closed form results for this integral as it depends on the marginal densities and the copula. However, we can additionally derive the following probability density functions which will prove useful in evaluating its properties:

$$f_{\omega|\pi}(\omega | \pi = c) = f_{\tilde{r}}(\omega - c) \partial_{1,2} \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)) \quad (\text{A26})$$

$$f_\omega(\omega) = \int_{-\infty}^{\infty} f_\pi(c) f_{\tilde{r}}(\omega - c) \partial_{1,2} \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)) dc \quad (\text{A27})$$

²Theorem 2.3 [Navarro and Sarabia 2022](#) shows that this admits a distortion function representation.

and the following corollary uses these representations to re-express Lemma 4 in terms of the joint distributions of π and $\tilde{r}$.

COROLLARY A1 (MLRP). *The monotone-likelihood ratio property for willingness-to-pay, ω , and cost type, π , is equivalent to the following property for the cost types and risk premia $\tilde{r}$:*

$$\frac{f_{\tilde{r}}(r-\pi)\partial_{1,2}\tilde{C}(F_{\pi}(\pi),F_{\tilde{r}}(r-\pi))}{f_{\tilde{r}}(r-\pi')\partial_{1,2}\tilde{C}(F_{\pi}(\pi'),F_{\tilde{r}}(r-\pi'))} \geq \frac{f_{\tilde{r}}(r'-\pi)\partial_{1,2}\tilde{C}(F_{\pi}(\pi),F_{\tilde{r}}(r'-\pi))}{f_{\tilde{r}}(r'-\pi')\partial_{1,2}\tilde{C}(F_{\pi}(\pi'),F_{\tilde{r}}(r'-\pi'))} \quad \forall \pi' \geq \pi \text{ and } r' \geq r$$

Thus, all joint distributions satisfying this property inherit properties (i) - (iii) in Proposition 4.

I now turn to discussing examples of common parametric copulas and the conditional distributions of ranks (summarized by their derivatives). While this allows for convenient theoretical analysis, it is worth emphasizing, however, that in empirical settings copulas can be estimated directly from the data using non-parametric tools. This is an inexhaustive list and other copula family also abound.^{3,4}

Example 1: Gaussian copula. The Gaussian copula is a family of copulas that vary in their correlation parameter ρ . All multivariate Normal distributions have Gaussian copulas and Normal marginals. However, not all normally distributed marginals have Gaussian copulas. The bivariate copula is defined as:

$$C(u, v; \rho) = \Phi^{-1}(u) \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\Phi^{-1}(v)} \exp \left[-\frac{1}{2} \frac{y^2 - 2\rho\Phi^{-1}(u)y + \Phi^{-1}(u)^2}{1-\rho^2} \right] dy \quad (\text{A28})$$

$$= \Phi^{-1}(u) \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{\Phi^{-1}(u)^2}{2} \right] \Phi \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A29})$$

with first derivative given by (it is symmetric)

$$\frac{d}{du}C(u, v; \rho) = \frac{d}{du}\Phi^{-1}(u) \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{\Phi^{-1}(u)^2}{2} \right] \Phi \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A30})$$

$$= \Phi \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right], \quad (\text{A31})$$

³Moreover, the convolution of copulas above is closed under mixtures which, with sufficient mixing, allows for almost arbitrary dependence using parametric families.

⁴A less common copula in statistical applications, but highly relevant in economic applications is the Burr copula. This is characterized by joint-Burr distributions which exhibit similar jointness properties to the normal distribution. Moreover, univariate Burr distributions have extensive use as a generalized version of the Pareto distribution to model income distributions with long tails (see, for instance, Singh and Maddala 1976).

and cross-partial derivative (or copula density)

$$\frac{\partial^2}{\partial u \partial v} C(u, v; \rho) = \frac{1}{\sqrt{1-\rho^2}} \frac{d}{dv} \Phi^{-1}(v) \Phi' \left[\frac{\Phi^{-1}(v) - \rho \Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A32})$$

$$= \frac{1}{\sqrt{1-\rho^2}} \exp \left[\frac{-\rho^2 \Phi^{-1}(v)^2 + 2\rho \Phi^{-1}(u) \Phi^{-1}(v) - \rho^2 \Phi^{-1}(u)^2}{2(1-\rho^2)} \right]. \quad (\text{A33})$$

Substituting into Equation A23 gives the following expression for the joint density in this example:

$$F_{\omega|\pi}(\omega|\pi = c) = \Phi \left[\frac{\Phi^{-1}(F_{\pi}(c)) - \rho \Phi^{-1}(F_{\tilde{r}}(\omega - c))}{\sqrt{1-\rho^2}} \right], \quad (\text{A34})$$

which highlights that the normality of the conditional density is a special property. It requires both the specific Gaussian copula and the normally distributed marginals.

Figure A3 shows how the parametric specification of the copula effects the joint distribution when the marginal distribution are fixed. The left columns shows how the joint distribution of π and $\tilde{r}$ changes with the correlation parameter. In both cases I assume π and $\tilde{r}$ are log normally distributed with their logs following standard normals. Comparing panels A and C shows the role of the correlation parameter for the gaussian copula. As the correlation increases, it stretches the density towards the upper right corner.

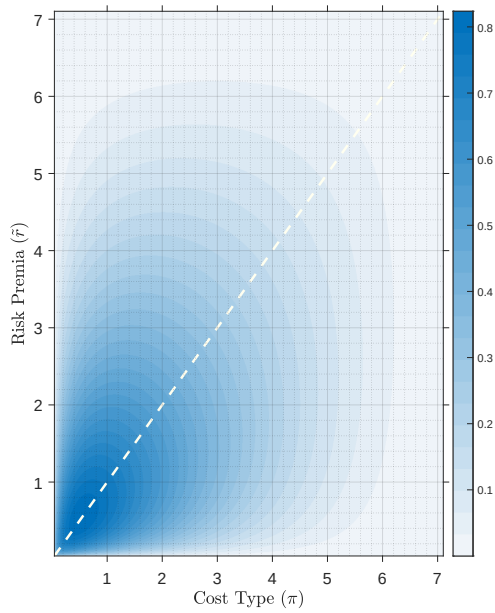
Example 2: Gumbel Copula. The Gumbel copula has heavier tails than the normal distribution and is also asymmetric, with more weight in the right tail. It's left tail, however, is similar to the normal distribution. The bivariate copula is defined (for $\delta \geq 1$) as:

$$C(u, v; \delta) = \exp \left(- \left[(-\log u)^\delta + (-\log v)^\delta \right]^{1/\delta} \right) \quad (\text{A35})$$

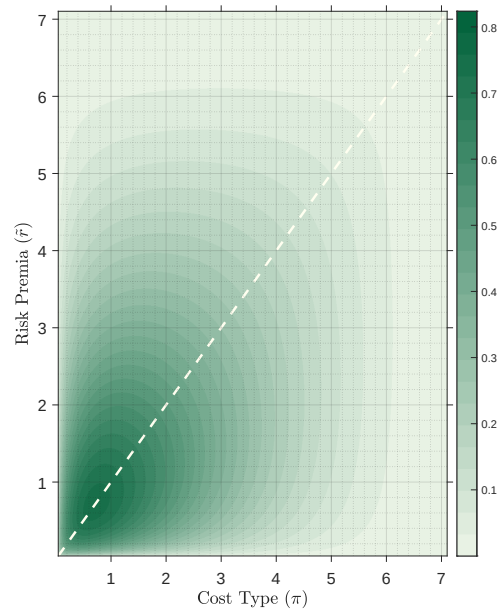
$$= \exp \left(-A^{1/\delta} \right) \quad \text{where} \quad A = (-\log u)^\delta + (-\log v)^\delta \quad (\text{A36})$$

with first derivative given by (it is symmetric)

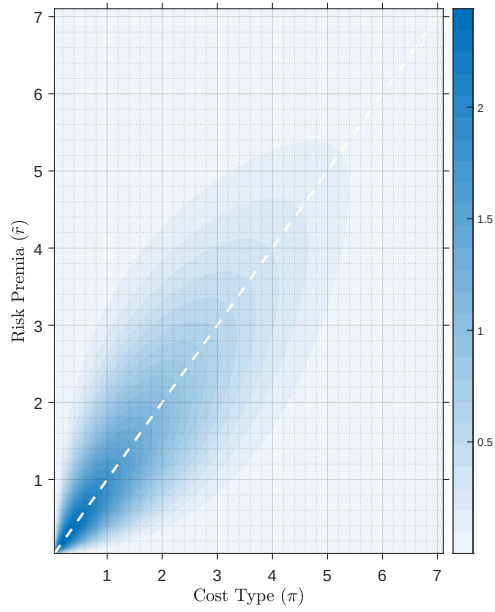
$$\frac{\partial}{\partial u} C(u, v; \delta) = \exp \left(-A^{1/\delta} \right) \cdot \frac{(1 + (-\log v / -\log u)^\delta)^{1/\delta-1}}{u} \quad (\text{A37})$$



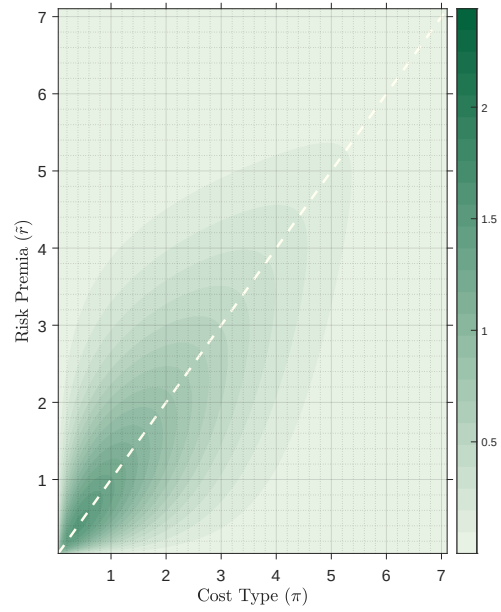
(A) Gaussian Copula (correlation=0.5)



(B) Gumbel Copula (correlation=0.5)



(C) Gaussian Copula (correlation=0.9)



(D) Gumbel Copula (correlation=0.9)

FIGURE A3. Copula and Joint Densities

Notes: Panels vary the copula (and parameter controlling correlation) for the rank dependency between cost type (π) and risk premia ($\tilde{r}$). In all panels the marginal distributions of cost type (π) and risk premia ($\tilde{r}$) are distributed log-normally with a mean of 0 and unit variance.

and cross-partial derivative

$$\frac{\partial^2}{\partial u \partial v} C(u, v; \delta) = ot \frac{(-\log v)^{\delta-1}}{v} \left[\frac{1-\delta}{\delta} (A^{1/\delta} + \delta - 1) A^{1/\delta-2} \right]. \quad (\text{A38})$$

Comparing the right column of Figure A3, which shows the Gumbel copula, to the left highlights differences induced in the joint density by choice of the copula. At lower correlations, panels A and B, the two copulas deliver more similar joint-densities. However, panel B, using the Gumbel copula, reveals heavier tails of the joint distribution. This is even more pronounced for higher correlations shown in panels C and D. Here, like the Gaussian copula, panel D stretches the density towards the upper-right corner of the figure. In contrast though, the normal exhibits softer contours. Instead, the surface of the joint density is pulled “tauter” under the Gumbel copula and more mass is concentrated in the joint tail. Thus the risk is more jointly heavily tailed.

Example 3: The Independent Copula. The independent copula describes the case where the two random variables are independent. The relevant formulae are:

$$C(u, v) = uv \quad (\text{A39})$$

$$\frac{\partial}{\partial u} C(u, v) = v \quad (\text{A40})$$

$$\frac{\partial^2}{\partial u \partial v} C(u, v) = 1. \quad (\text{A41})$$

Appendix E. General Proofs and Additional Results for Sections 4.2 and 4.4

E.1. Additional Results

I begin by proving demand contraction under the following alternative assumptions:

ASSUMPTION A1. *The perceived risk premia (net of any behavioral influences) of insurance contracts are lower for immature markets. $\tilde{r}^{MM} \succeq \tilde{r}^{IM}$*

ASSUMPTION A2. *Risk types, π_i , have identical marginal distributions in mature and immature markets.*

and

ASSUMPTION A3. *Resorting satisfies the following two conditions for the copulas C , the bivariate copula denoting the dependence between π_i and $\tilde{r}_i$, and $\hat{C}$:*

- i. *The partial derivatives of $\tilde{C}$ satisfy $\frac{\partial \tilde{C}^{MM}(u, F_{\tilde{r}}^{MM}(\tilde{r}))}{\partial u} \leq \frac{\partial \tilde{C}^{IM}(u, F_{\tilde{r}}^{IM}(\tilde{r}))}{\partial u} \quad \forall u \in [0, 1] \text{ and } \tilde{r}.$*

PROPOSITION A1 (Demand Contraction). *If $\omega_i = u(\pi_i + \tilde{r}_i)$, willingness-to-pay is on average larger and demand expands in mature markets. $D^{IM}(p) \leq D^{MM}(p) \quad \forall p$*

PROOF OF PROPOSITION A1. First consider linear utility with $\omega_i = \pi_i^{IM} + \tilde{r}_i^{IM}$. Assumption A3 and Assumption A1 guarantee that the conditions in Theorem 3.3 in Navarro and Sarabia (2022) are satisfied. A direct application gives

$$\omega_i^{IM} = \pi_i^{IM} + r_i^{IM} + \varepsilon_i^{IM} \preceq \pi_i^{MM} + r_i^{MM} + \varepsilon_i^{MM} = \omega_i^{MM} \longrightarrow F_{\omega}^{MM}(p) \leq F_{\omega}^{IM}(p) \quad \forall p. \quad (\text{A42})$$

As the usual stochastic order is preserved under increasing transformations, the stated proposition then follows immediately for any increasing $u(\cdot)$ $\square$

This level effect does not change the identity of marginal (and inframarginal) consumers at any quantity demanded is the same and the equilibrium price increases and demand falls (See Proposition 2).

Moreover, when there is *only* a level effect the following lemma shows that the utility function can be generalized to allow for arbitrary substitution patterns between π_i .

LEMMA A1 (Arbitrary Substitutes). *When there is no re-sorting ($\tilde{C}^{IM} = \tilde{C}^{MM}$), Proposition A1 can be extended to any willingness-to-pay satisfying*

$$\omega_i = u(\pi_i, \tilde{r}_i) \quad (\text{A43})$$

where $u(\cdot, \cdot)$ is a function increasing in its arguments.

PROOF OF LEMMA A1. Identical in distribution implies $\pi^{MM} \succeq \pi^{IM}$. Thus, each element of the vector $(\pi_i, \tilde{r}_i)$ in mature markets is larger in the usual stochastic order than in immature markets. It follows that the vector $\mathbf{X}^{IM} \preceq \mathbf{X}^{MM}$ in the usual multivariate stochastic ordering (Shaked and Shanthikumar 2007, Theorem 6.B.14).

This order is preserved for any increasing function $u : \mathbb{R}^2 \rightarrow \mathbb{R}$, then $u(\mathbf{X}^{IM}) \preceq u(\mathbf{X}^{MM})$. Thus, $F_\omega^{IM}(p) \geq F_\omega^{MM}(p) \forall p$ and the stated lemma follows. This result extends to any multi-dimensional vector (for example r and ε) under corresponding assumptions on their marginals. $\square$

For the special case of the normal distribution, demand contractions are tightly linked to the price elasticity of consumers in immature and mature markets. The following corollary formalises this for the general case:

COROLLARY A2 (Price Elasticity of Demand). *When willingness-to-pay is smaller in the hazard rate order in mature markets, that is*

$$\frac{f^{IM}(\omega)}{1 - F^{IM}(\omega)} \geq \frac{f^{MM}(\omega)}{1 - F^{MM}(\omega)} \forall p, \quad (\text{A44})$$

then demand is more price sensitive in immature markets.

PROOF. This follows from substituting the definition of demand into the usual formula for the elasticity,

$$\eta_{D,p} = D'(p) \frac{p}{D(p)} = -f(p) \frac{p}{1 - F(p)}, \quad (\text{A45})$$

which implies in the normal case

$$\eta_{D,p}^{IM} - \eta_{D,p}^{MM} \propto - \underbrace{\left(\frac{\phi(\frac{p-\mu_\omega^{IM}}{\sigma_\omega})}{1 - \Phi(\frac{p-\mu_\omega^{IM}}{\sigma_\omega})} - \frac{\phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})}{1 - \Phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})} \right)}_{\text{level effect} \geq 0} \leq 0 \quad (\text{A46})$$

The inverse Mills ratio and the difference in the inverse Mills ratio are always positive by the log concavity of the normal distribution ensuring non-negativity. Thus consumers are more price elastic in immature markets. $\square$

E.2. Proofs

PROOF OF LEMMA 1. Analogously to the proof of Proposition 4

$$F_{\pi|\omega}^{IM}(\pi|\omega) \succeq F_{\pi|\omega}^{MM}(\pi|\omega) \longrightarrow MC^{IM}(p) \geq MC^{MM}(p) \forall p. \quad (A47)$$

As the unconditional first order stochastic dominance of private valuations in mature and immature markets implies $F_{\omega}^{MM}(\omega) \leq F_{\omega}^{IM}(\omega)$, (A47) holds if and only if the copulas satisfy

$$C_2^{MM}(u, a) \leq C_2^{IM}(u, b) \forall a \geq b \text{ and } \forall u. \quad (A48)$$

This is trivially satisfied under identical concave copulas ($C = C^{IM} = C^{MM}$). However, it cannot be satisfied for different copulas that are both concave in their second argument (see Proposition 4, condition(i)). $\square$

PROOF OF PROPOSITION 5. The proof in the general case follows from a direct application of Proposition 2.1 in Fang and Joe (1992). $\square$

PROOF OF PROPOSITION 6. Recall the following necessary and sufficient condition for the comparison of average costs (See the proof of Proposition 4):

$$F_{\pi|\omega \geq p}(\pi|\omega \geq v_1) \leq F_{\pi|\omega \geq p}(\pi|\omega \geq v_2) \forall v_1 \geq v_2 \text{ and } \forall \pi \quad (A49)$$

$$\frac{u - C(u, F_{\omega}(v_1))}{1 - F_{\omega}(v_1)} \leq \frac{u - C(u, F_{\omega}(v_2))}{1 - F_{\omega}(v_2)} \forall v_1 \geq v_2 \text{ and } \forall u \quad (A50)$$

The analogous condition to order average costs between markets is:

$$F_{\pi|\omega \geq p}^{IM}(\pi|\omega \geq v) \leq F_{\pi|\omega \geq p}^{MM}(\pi|\omega \geq v) \forall v, \pi \quad (A51)$$

$$\frac{u - C^{IM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{MM}(v))}{1 - F_{\omega}^{MM}(v)} \forall v, \forall u \quad (A52)$$

As $F_{\omega}^{MM}(\omega) \leq F_{\omega}^{IM}(\omega)$, established in Proposition A1, the following is true for all copulas satisfying condition (ii) of Proposition 4 (see the proof above)

$$\frac{u - C^{MM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{MM}(v))}{1 - F_{\omega}^{MM}(v)} \forall v \text{ and } \forall u, \quad (A53)$$

and by the concordance ordering in Assumption A3 we have

$$\frac{u - C^{IM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \quad \forall v \text{ and } u \in [0, 1], \quad (\text{A54})$$

which completes the proof. $\square$

PROOF OF PROPOSITION 7. Note that the copula defines *rank* dependence. Consequently, they are invariant under strictly monotone transformations, including the inverse demand transformation considered here. Thus, to order average costs given quantity, q , we have the following condition:

$$\frac{u - C^{IM}(u, q)}{1 - q} \leq \frac{u - C^{MM}(u, q)}{1 - q} \quad \forall q \in [0, 1] \text{ and } u \in [0, 1], \quad (\text{A55})$$

which simplifies to

$$C^{IM}(u, q) \geq C^{MM}(u, q) \quad \forall q \in [0, 1] \text{ and } u \in [0, 1]. \quad (\text{A56})$$

This only depends on the change in sorting ($C^{IM} \neq C^{MM}$) and holds with equality under no re-sorting ($C = C^{IM} = C^{MM}$). The stated Proposition then follows from the concordance ordering in Assumption A3.

Equation (A48) gives an identical condition for marginal costs in terms of the conditional density which corresponds to stochastic dominance

$$C_2^{IM}(u, q) \geq C_2^{MM}(u, q) \quad \forall q \in [0, 1] \text{ and } u \in [0, 1]. \quad (\text{A57})$$

This cannot be satisfied as stochastic dominance for all conditional distributions implies stochastic dominance of the unconditional distribution. This violates assumption 2.

Note that at $q = 0$ marginal costs are equal to average costs. Then, the monotonicity of marginal costs and the results of Proposition 5 imply a single crossing property which completes the proof of rotation.

The rotation of the marginal cost curve and Proposition 5 imply the derivative of the marginal cost curve is also ordered in quantity space

$$\frac{\partial MC^{IM}(q)}{\partial p} \leq \frac{\partial MC^{MM}(q)}{\partial q} < 0, \quad (\text{A58})$$

and we can express the wedge as

$$W(q) = \frac{1}{q} \int_0^q (MC(t) - MC(q)) dt = \int_0^q \left(MC'(t) \frac{t}{q} \right) dt. \quad (\text{A59})$$

Combining them gives $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$ and $AC^{IM}(q) - MC^{IM}(q) \leq AC^{MM}(q) - MC^{MM}(q) < 0$. □

PROOF OF LEMMA 2. We omit the subscript j for convenience. Let $S(q)$ denote the supply schedule in a market,

$$S(q) = MC(q) + \frac{1}{\eta_D(q)} \quad (\text{A60})$$

Excess supply is given by $ES(q) = p(q) - S(q)$ and $ES^{MM}(q^{MM}) = 0$ by definition. Then

$$ES^{IM}(q^{MM}) = p^{IM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM}) \right) \quad (\text{A61})$$

$$\leq p^{MM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM}) \right), \quad (\text{A62})$$

by the assumption of the demand contraction. Under the restriction on quantities and elasticities stated in the lemma we have

$$p^{MM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM}) \right) \quad (\text{A63})$$

$$= \underbrace{p^{MM}(q^{MM}) - \left(MC^{MM}(q^{MM}) + 1/\eta_D^{MM}(q^{MM}) \right)}_{=ES^{MM}(q^{MM})=0} \quad (\text{A64})$$

$$+ \underbrace{\left(MC^{MM}(q^{MM}) - MC^{IM}(q^{MM}) \right)}_{\leq 0} + \underbrace{\left(1/\eta_D^{MM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM}) \right)}_{\leq -(MC^{MM}(q^{MM}) - MC^{IM}(q^{MM}))} \leq 0,$$

which establishes excess supply in immature markets at the mature market equilibrium. The remainder of the proof follows analogously to the proof of Proposition 2. Thus $q_j^{MM} \geq q_j^{IM}$. It follows that $p_j^{MM} \leq p_j^{IM}$ by the rotation of the marginal cost curve and restriction on elasticities. □

References

- Cherubini, U., S. Mulinacci, F. Gobbi, and S. Romagnoli (2011). *Dynamic Copula Methods in Finance*. John Wiley & Sons.
- Fang, Z. and H. Joe (1992). Further developments on some dependence orderings for continuous bivariate distributions. *Annals of the Institute of Statistical Mathematics* 44, 501–517.
- Mitchell, O. S., J. M. Poterba, M. J. Warshawsky, and J. R. Brown (1999). New evidence on the money's worth of individual annuities. *American Economic Review* 89(5), 1299–1318.
- Navarro, J. and J. M. Sarabia (2022). Copula representations for the sum of dependent risks: models and comparisons. *Probability in the Engineering and Informational Sciences* 36(2), 320–340.
- Nelsen, R. B. (2006). *An Introduction to Copulas*. Springer.
- Poterba, J. M. and A. Solomon (2021). Discount rates, mortality projections, and money's worth calculations for us individual annuities. Technical report, National Bureau of Economic Research.
- Shaked, M. and J. G. Shanthikumar (2007). *Stochastic orders*. Springer.
- Singh, S. K. and G. S. Maddala (1976). A function for size distribution of incomes. *Econometrica* 44(5), 963–970.